



OECD Statistics Working Papers 2026/03

OECD methodology
to estimate SEEA Air
Emission Accounts: An
update

**Santaro Sakata,
Suyeon Hwang,
Roberto Astolfi,
Bram Edens**

<https://dx.doi.org/10.1787/6fc092d8-en>

OECD methodology to estimate SEEA Air Emission Accounts: an update

By Santaro Sakata, Suyeon Hwang, Roberto Astolfi and Bram Edens

OECD Statistics Working Papers

[See all the papers in the series](#)

The OECD Statistics Working Paper Series – managed by the OECD Statistics and Data Directorate – is designed to make available in a timely fashion and to a wider readership selected studies prepared by OECD staff or by outside consultants working on OECD projects. The papers included are of a technical, methodological or statistical policy nature and relate to statistical work relevant to the Organisation. The Working Papers are generally available only in their original language – English or French – with a summary in the other.

This work is published under the responsibility of the Secretary-General of the OECD. The opinions expressed and arguments employed herein do not necessarily reflect the official views of the Member countries of the OECD.

Working Papers describe preliminary results or research in progress by the authors and are published to stimulate discussion on a broad range of issues on which the OECD works. Comments on Working Papers are welcomed and may be sent to the Statistics and Data Directorate, OECD, 2 rue André Pascal, 75775 Paris Cedex 16, France or by email to stat.contact@oecd.org.

This document, as well as any statistical data and map included herein, are without prejudice to the status of or sovereignty over any territory, to the delimitation of international frontiers and boundaries and to the name of any territory, city or area.

The release of this working paper has been authorised by Steve MacFeely, OECD Chief Statistician and Director of the OECD Statistics and Data Directorate.

© OECD (2026)



Attribution 4.0 International (CC BY 4.0)

This work is made available under the Creative Commons Attribution 4.0 International licence. By using this work, you accept to be bound by the terms of this licence <https://creativecommons.org/licenses/by/4.0/>.

Attribution – you must cite the work.

Translations – you must cite the original work, identify changes to the original and add the following text: *In the event of any discrepancy between the original work and the translation, only the text of original work should be considered valid.*

Adaptations – you must cite the original work and add the following text: *This is an adaptation of an original work by the OECD. The opinions expressed and arguments employed in this adaptation should not be reported as representing the official views of the OECD or of its Member countries.*

Third-party material – the licence does not apply to third-party material in the work. If using such material, you are responsible for obtaining permission from the third party and for any claims of infringement.

You must not use the OECD logo, visual identity or cover image without express permission or suggest the OECD endorses your use of the work.

Any dispute arising under this licence shall be settled by arbitration in accordance with the Permanent Court of Arbitration (PCA) Arbitration Rules 2012. The seat of arbitration shall be Paris (France). The number of arbitrators shall be one.

Abstract

This working paper updates the existing OECD methodology for estimating Air Emission Accounts (AEAs) to enhance their comprehensiveness and international comparability. The new methodology draws primarily on data from national greenhouse gas emission inventories, a range of auxiliary data sources including Physical Energy Flow Accounts, as well as OECD experimental estimates of emissions from air and maritime transport.

It improves the quality and granularity of estimates by economic activity (A*64 industry breakdown), expands the geographical coverage to all OECD countries, and extends the temporal scope of the estimates (1990 to year $t-2$). It also widens the emission coverage to include international air and maritime transport in line with emission boundaries of the System of Environmental Economic Accounting (SEEA), and incorporates the territory–residence adjustments to comply with the SEEA Central Framework (SEEA-CF).

These improvements complement ongoing international methodological efforts to refine AEA estimates, such as those developed by the IMF and Eurostat, by drawing more extensively on country-specific data and achieving closer alignment with the SEEA-CF.

JEL codes: C82, E01, Q53, Q54, Q56.

Résumé

Ce document de travail actualise la méthodologie existante de l'OCDE pour l'estimation des comptes d'émissions atmosphériques (AEAs) afin d'en améliorer l'exhaustivité et la comparabilité internationale. La nouvelle méthodologie s'appuie principalement sur les données issues des inventaires nationaux d'émissions de gaz à effet de serre, sur un ensemble de sources de données auxiliaires, notamment les comptes des flux physiques d'énergie, ainsi que sur les estimations expérimentales de l'OCDE relatives aux émissions du transport aérien et maritime.

Elle améliore la qualité et le niveau de détail des estimations par activité économique (ventilation des branches d'activité A*64), étend la couverture géographique à l'ensemble des pays de l'OCDE et élargit la couverture temporelle des estimations (de 1990 à l'année $t - 2$). Elle étend également le champ des émissions afin d'inclure le transport aérien et maritime international, conformément aux frontières d'émissions définies par le Système de comptabilité économique et environnementale (SCEE), et intègre les ajustements entre les principes du territoire et de la résidence afin d'assurer la conformité avec le Cadre central du SCEE.

Ces améliorations complètent les travaux méthodologiques internationaux en cours visant à affiner les estimations des comptes d'émissions atmosphériques, notamment ceux menés par le FMI et Eurostat, en recourant plus largement à des données propres à chaque pays et en renforçant l'alignement sur le Cadre central du SCEE.

Classification JEL : C82, E01, Q53, Q54, Q56.

Acknowledgements

This paper was drafted by Santaro Sakata, Suyeon Hwang, Roberto Astolfi, and Bram Edens from the Statistics and Data Directorate. The authors would like to express their particular gratitude to Manon Roth for her substantial contributions to the project, including the analysis of correspondence tables and the development of the foundational code used to estimate AEAs.

The work was carried out under the supervision of Jorrit Zwijnenburg, Head of Division of the National Accounts Division at the OECD Statistics and Data Directorate.

The authors are particularly grateful to Amanda Penistone (Statistics and Data Directorate) for her thorough review and insightful comments on drafts of this paper. The authors would also like to thank various OECD colleagues for their helpful comments during discussions or on earlier versions of this paper, including Karine Tremblay, Matthew de Queljoe and Pearl Herrero (Statistics and Data Directorate), as well as Norihiko Yamano and Michel Lioussis (Directorate for Science, Technology and Innovation).

The authors are grateful to Australia and Korea for their exchanges and feedback on the updated methodology and/or preliminary results. The authors thank country representatives and international organisations for their review of earlier versions of this paper and for their helpful comments and constructive suggestions provided through the Joint UNECE/OECD Expert Meeting on Implementation of SEEA (March 2026), the OECD Working Party on Environmental Information (WPEI) (March 2026), and the SEEA Central Framework Technical Committee (May 2026).

Table of contents

OECD Statistics Working Papers	2
Acknowledgements	5
1 Introduction	8
2 Updated methodology	10
General approach	10
Allocation	13
Air and maritime transport emissions	17
3 Data sources	18
Inventories	18
Auxiliary data sources	20
OECD estimates on CO ₂ emissions from air and maritime transport	23
4 Preliminary results	24
5 Discussion and plausibility checks	27
Key improvements	27
Plausibility checks: Approach	27
Plausibility checks: Results	31
6 Dissemination Strategy	41
7 Next steps	43
References	45
Annex A. Correspondence table	50
Annex B. Distribution of misallocated emissions excluding air transport, maritime/fishing, and road transport	56
Annex C. Plausibility of allocation of road transport emissions	58
Notes	61

FIGURES

Figure 1. CO ₂ emissions with industry/household breakdown	24
Figure 2. Total CO ₂ emissions: AEA, CRT/PRIMAP and IEA	25
Figure 3. Industry and household CO ₂ emissions share in 2022	26
Figure 4. Comparison between OECD AEA estimates and official AEAs, Country Group 1, 2014-2023	31
Figure 5. Mean Absolute Percentage Error (MAPE) of total CO ₂ emissions (excluding air transport, maritime/fishing and road transport), Country Group 1, 2014-2023	33
Figure 6. Correlation coefficients of growth rates between official and estimated AEAs, 2014-2023	34
Figure 7. Mean Absolute Percentage Error (MAPE) of road transport emissions, Country Group 1, 2014-2023	35
Figure 8. Mean Absolute Percentage Error (MAPE) of total CO ₂ emissions, Country Group 2, 2014-2023	36
Figure 9. Misallocation share of total CO ₂ emissions (excluding air transport, maritime/fishing, and road transport), Country Group 1, 2014-2023	37
Figure 10. Misallocation share by level of industry breakdown and allocation basis (excluding air transport, maritime/fishing, and road transport), Country Group 1, 2014-2023	38
Figure 11. Comparison of misallocation shares under the 2018 and updated methodology, Country Group 1, 2014-2023	39
Figure 12. Misallocation share, Country Group 2, 2014-2023	40
Figure 13. Comparison of industry and household emissions between OECD CO ₂ estimates and official GHG total of Australia, 2004/05-2016/17	40
Figure 14. Industry and household distribution of misallocated emissions (excluding air transport, maritime/fishing and road transport), Country Group 1	57
Figure 15. Misallocation share of road transport CO ₂ emissions, Country Group 1	58
Figure 16. Misallocation shares, across different auxiliary data	59
Figure 17. Industry and household distribution of misallocated road transport emissions, Country Group 1	60

TABLES

Table 1. CRT and ISIC classification systems	11
Table 2. Availability of inventory data after 1990	19
Table 3. Energy use data availability at the OECD/IEA Data Explorer after 1990	21
Table 4. Data availability of STAN Database for Structural analysis after 1990	22
Table 5. Data availability on supply tables	22
Table 6. Availability of official CO ₂ AEAs	28
Table 7. Industry breakdown for misallocation assessment	30
Table A.1. Information related to correspondence between inventory categories and economic and household activities	51
Table A.2. Reference Correspondence Table for CO ₂ emissions	53

1 Introduction

1. Building on the 2018 OECD methodology (Flachenecker, Guidetti and Pionnier, 2018^[1]), this working paper presents an updated approach for estimating Air Emission Accounts (AEAs) in line with the System of Environmental-Economic Accounting (SEEA). The updated methodology improves upon the existing 2018 approach by: (1) enhancing quality and granularity of estimates by economic activity (industry) incorporating additional data sources, (2) extending geographical and temporal coverage; (3) including previously excluded international transport emissions and territory-residence adjustments for mobile sources of emissions to comply with the SEEA Central Framework (SEEA-CF). Together, these improvements strengthen conceptual consistency with SEEA principles, international comparability, and the policy relevance of the resulting AEA estimates. At present, the gas coverage is limited to carbon dioxide (CO₂), and no estimates of total greenhouse gas (GHG) emissions are included. However, the methodology is designed to allow future extensions to other GHGs.

2. This work contributes to international efforts to expand and harmonise the availability of AEA estimates by improving the existing methodology to estimate the AEAs for countries that do not regularly compile official AEAs. Relevant initiatives to date include: (i) the OECD methodology for estimating AEAs (Flachenecker, Guidetti and Pionnier, 2018^[1]), which this paper builds upon and updates; (ii) production-based emissions estimates by economic activity developed by the OECD Directorate for Science, Technology and Innovation (STI) in the framework of carbon-footprint calculations (Yamano, Lioussis and Cimper, 2024^[2]); (iii) Eurostat's AEA estimates for non-European countries used as inputs to their footprint analysis (Eurostat, 2025^[3]); and (iv) the AEA estimation tool developed by the International Monetary Fund (IMF) within the context of the G20 Data Gaps Initiative (DGI) (IMF, 2025^[4]).

3. While the 2018 OECD methodology represented a milestone in filling gaps for countries not compiling AEAs, it exhibited several limitations:

- its geographical coverage was restricted to United Nations Framework Convention on Climate Change (UNFCCC) Annex I countries¹,
- GHG coverage excluded fluorinated greenhouse gases,
- time coverage and industry disaggregation was constrained by the availability of official inventory and output data,
- estimates were not fully aligned with SEEA boundaries, notably omitting international air and maritime transport,
- it included no adjustments from territory- to residence-principles (see Box 1),
- output data were the sole auxiliary information used to allocate inventories to industries, and
- road-transport allocations relied on allocation ratios derived from a narrow sample of European countries.²

4. Similarly, the IMF and Eurostat estimation approaches do not implement territory–residence adjustments and rely on unvarying parameters across countries and/or years. The present methodology update addresses these limitations.

5. Despite the ongoing international efforts to expand the AEA coverage, including the launch of a global data collection by the OECD and the UN Statistics Division in 2023, major gaps in the availability of

official AEAs persist, although it is mandatory to compile AEAs in the European Union (European Union, 2011^[5]). According to the 2025 Global Assessment conducted by the United Nations, only 53 countries have compiled AEAs over the past five years, including non-disseminated pilot compilations (UNCEEA, 2026^[6]). Beyond the limited geographical coverage, important differences in gas scope, time-series length, and level of detail by industry further constrain the comparability of the AEAs and hence their analytical and policy use. The updated methodology helps broaden the availability of information by providing granular estimates with the A*64 industry breakdown, a standard aggregation of economic activities into 64 industries (European Union, 2010^[7]), and extending the temporal scope to the period from 1990 to year $t-2$. It also improves international comparability by aligning with the emissions coverage and principles of the SEEA. It is in principle suitable for all OECD countries, owing to the availability of the required input data. In this research, however, it has been applied to non-EU OECD countries that do not currently report official AEA estimates, and to European countries to support plausibility checks.

6. The resulting AEA estimates can support a wide range of analytical and policy applications. They enable emissions to be analysed within the framework and perimeter of national accounts, i.e. alongside macroeconomic variables such as gross domestic product (GDP), value added, and output. This facilitates assessments of emissions intensity and thus analysis of decoupling between environmental pressures and economic growth. When combined with macroeconomic projections, AEAs can also support estimates of future emissions and monitoring of progress towards emission-reduction targets. Finally, AEAs provide essential inputs for carbon-footprint calculations.

7. This paper is organised as follows. Section 2 describes the updated methodology for estimating AEAs. Section 3 outlines the data sources used and explains how time series gaps are addressed. Section 4 presents preliminary results, and Section 5 assesses their plausibility across different country groups. Section 6 discusses a possible dissemination strategy. Section 7 concludes with next steps. The Annexes provide additional technical detail, including the correspondence table used to allocate emissions from inventories to economic activities (industries) and households, and detailed plausibility checks.

2 Updated methodology

General approach

8. The proposed methodology update maintains the widely applied practice of allocating national inventory emissions to economic activities (classified by the International Standard Industrial Classification of All Economic Activities (ISIC) or its regional equivalents) and households, the so-called “inventory-first” approach (Eurostat, 2015^[8]) (see Box 1 for key differences between emission inventories and accounts). This approach helps preserve consistency between emission inventories and accounts and is consistent with the 2018 OECD methodology, as well as with methods used by the IMF, Eurostat, and many national statistical offices. It also offers practical advantages for extending the geographical coverage of AEA estimates compared with the alternative “energy-first” approach, given the limited availability of Physical Energy Flow Accounts (PEFA) data (see Section 3).

9. Under this approach, national inventory data³ serve as the starting point, but several adjustments are made. First, system boundaries are adjusted to go from the territory to the residence principle by deducting emissions generated by non-residents on the territory (e.g. domestic aviation by non-resident airlines bunkering at airports of the reporting country) and adding emissions from international transport by residents as well as domestic transport by residents fuelling abroad. Hereto, OECD experimental estimates of CO₂ emissions from air and maritime transport are applied. Second, GHG inventory emission source categories are mapped to ISIC economic activities and households using a correspondence table. Where an emission source corresponds to a single ISIC economic activity or households in the correspondence table, emissions are allocated directly to that category (“1-to-1” mapping). Where an emission source corresponds to multiple ISIC economic activities or households (“1-to-many” mapping), emissions from the inventory category are distributed across the relevant ISIC activities and households. To do so, allocation coefficients (weights) across multiple ISIC economic activities or households are derived based on various auxiliary data sources following a preference order, as discussed later.

10. When inventory emissions data are available for both parent and child industry categories, emissions reported in the more detailed child categories are used for mapping, unless this results in implausible or incomplete industry allocation⁴, or the sum of the child categories does not match the parent category. In the absence of child-category data, emissions are mapped from the parent category.

11. The allocation can be expressed as a matrix multiplication, where emissions by ISIC economic activities and households are obtained by multiplying a matrix of allocation coefficients by a vector of emissions reported in inventory categories:

$$a = Wc,$$

where a is the resulting vector of emissions by ISIC economic activity and households, W is the allocation (weight) matrix, and c is the vector of emissions by inventory category⁵. The entries in each column of the allocation matrix W represent how emissions by inventory categories are distributed among mutually exclusive ISIC economic activities and households. If inventory categories are in a one-to-one correspondence with an ISIC or households category, then the corresponding columns of the matrix W would consist of a 1 in the i -th position identifying the ISIC category and zeros elsewhere (i.e. a unit vector).

Otherwise, each column comprises multiple positive entries, each representing the proportion of emissions attributable to the respective ISIC categories. In either case, the sum of the entries in each column of W equals 1.

12. As a result, each entry given by the product $w_{j,k} \cdot c_k$ represents the quantity of emissions from inventory category k allocated to ISIC activity or households category j . Therefore, the proposed approach not only enables bridging GHG emission inventories based on the IPCC Guidelines to SEEA AEs but also expands the information set by providing a more comprehensive and granular data set.

13. The main challenges lie in the length and the level of granularity of the IPCC inventory series, as well as the auxiliary information required to estimate the proportions by which inventory categories are allocated to ISIC economic activities and households (see Section 3 for data availability of inventory data and extension of time series).

14. This section examines the following aspects of the methodology in greater detail: (i) the use of inventory data (i.e. the vector c in the equation above) as the starting point for emissions allocation, (ii) the allocation of inventory emissions to economic activities and households (i.e. multiplication by the matrix W), and (iii) the treatment of air and maritime transport emissions (including fishing), which are handled separately from the allocation of inventory emissions.

Box 1. Key differences between emission inventories and accounts

Emissions source classification systems

National GHG emissions inventories are reported by countries to the UNFCCC. The presented methodology uses national inventory data reported through the Common Reporting Table (CRT) as the primary source (see Section 3). Emissions inventories are classified by source categories and subcategories in accordance with the Intergovernmental Panel on Climate Change (IPCC) Guidelines. Table 1 presents the two-digit-level of emission source categories used in the CRT, alongside letter-level Sections in the ISIC classification system applied in AEs.

Table 1. CRT and ISIC classification systems

Sectors in CRT	Sections in the ISIC, Rev.4
1. Energy	A. Agriculture, forestry and fishing
1.A Fuel Combustion	B. Mining and quarrying
1.B Fugitive Emissions from Fuels	C. Manufacturing
1.C CO ₂ Transport and Storage	D. Electricity, gas, steam and air conditioning supply
2. Industrial Processes and Product Use (IPPU)	E. Water supply; sewerage, waste management and remediation activities
2.A Mineral Industry	F. Construction
2.B Chemical Industry	G. Wholesale and retail trade; repair of motor vehicles and motorcycles
2.C Metal Industry	H. Transportation and storage
2.D Non-energy Products from Fuels and Solvent Use	I. Accommodation and food service activities
2.E Electronics Industry	J. Information and communication
2.F Product Uses as Substitutes for ODS	K. Financial and insurance activities
2.G Other Product Manufacture and Use	L. Real estate activities
2.H Other	M. Professional, scientific, and technical activities
3. Agriculture	N. Administrative and support service activities
3.A Enteric Fermentation	O. Public administration and defence; compulsory social security
3.B Manure Management	P. Education

3.C Rice Cultivation	Q. Human health and social work activities
3.D Agricultural Soils	R. Arts, entertainment and recreation
3.E Prescribed Burning of Savannas	S. Other service activities
3.F Field Burning of Agricultural Residues	T. Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use
3.G Liming	U. Activities of extraterritorial organizations and bodies
3.H Urea Application	
3.I Other Carbon-containing Fertilizers	
3.J Other	
4. Land Use, Land Use Change and Forestry (LULUCF)	
4.A Forest Land	
4.B Cropland	
4.C Grassland	
4.D Wetlands	
4.E Settlements	
4.F Other Land	
4.G Harvested Wood Products	
4.H Other	
5. Waste	
5.A Solid Waste Disposal	
5.B Biological Treatment of Solid Waste	
5.C Incineration and Open Burning of Waste	
5.D Wastewater Treatment and Discharge	
5.E Other	
6. Other	

Note: The CRT categories and ISIC economic activities are presented in alphanumeric order in each classification system. Categories appearing in the same row do not correspond to one another.

Source: (UNFCCC, 2021^[9]; United Nations, 2008^[10])

IPCC source categories refer to emissions generated by specific physical sources and processes. In contrast, ISIC classifies productive activities based on the inputs of goods, services, and factors of production; the process and technology of production; the characteristics of outputs; and the use to which the outputs are put (United Nations, 2008^[10]). For instance, IPCC emissions category 3 “Agriculture” includes subcategory 3.A “Enteric fermentation” which describes a biological process. By contrast, ISIC Section A “Agriculture, forestry and fishing” is disaggregated into Division 01 “Crop and animal production, hunting and related service activities”, and further into Group 014 “Animal production” (Sakata, Aklilu and Pizarro, 2024^[11]).

Importantly, ISIC classification is based on the principal activity of the firm. For example, emissions from fuel use in agricultural transport are classified under CRT category 1 “Energy” instead of CRT category 3 “Agriculture”. In AEAs, however, these emissions are allocated to ISIC Section A “Agriculture, forestry and fishing” if the principal activity is agricultural production. Emissions are allocated to ISIC Section H “Transportation and storage” only when the principal activity of the firm is the provision of transport services (Flachenecker, Guidetti and Pionnier, 2018^[1]; Sakata, Aklilu and Pizarro, 2024^[11]).

Territory and residence principles

GHG emission inventories and AEAs also differ in their underlying accounting principles. Emission inventories follow the territory principle, whereby emissions are attributed to the geographical area in which they occur, regardless of the residency of the emitting entity. This means that all emissions

generated within a country's borders are included, including those from non-resident units operating on the territory.

In contrast, AEAs are compiled according to the residence principle, consistent with the System of National Accounts (SNA). Under this approach, emissions are attributed to the resident units of an economy, irrespective of where the emissions physically take place. As a result, emissions generated abroad by resident units (e.g. international transport operated by resident companies) are included, while emissions generated domestically by non-residents are excluded. These differences are particularly relevant for activities such as international transport and tourism, where the attribution of emissions can differ significantly between the two accounting frameworks (Sakata, Aklilu and Pizarro, 2024^[11]).

Scope difference of national totals

In national GHG inventories, emissions from international aviation and marine bunkers, as well as CO₂ emissions from biomass combustion, are reported under memo items (information items) and are excluded from national emissions totals (IPCC, 2006^[12]). In the case of CO₂ emissions from biomass combustion, this is to avoid double counting, since these emissions are reported in the LULUCF sector. In contrast, emissions from international transport by resident units are included in national totals in AEAs, and it is recommended that CO₂ emissions from biomass combustion be recorded separately from emissions arising from the combustion of fossil fuels (United Nations, 2014^[13]).

In this paper, emissions from international transport by resident units are incorporated directly from the OECD experimental estimates, as described further below. Stationary emissions are allocated from the national inventory categories, excluding CO₂ emissions from biomass combustion.

Allocation

Core allocation strategy

15. The correspondence from CRT categories to ISIC economic activities and households is presented in the OECD reference correspondence table in Annex A. This table excludes the LULUCF category to align with the AEA emission scope under the current SEEA-CF. It supports the A*64 industry breakdown, ensuring comparability between the resulting estimated AEAs and official AEAs, as the AEA questionnaires of Eurostat and the OECD/UNSD follow this level of detail. Annex A also summarises its development and reviews the main correspondence tables examined.

16. For 1-to-many mappings, inventory emissions are distributed across industries and households using allocation coefficients (weights). When a country reports all detailed emission source categories for CO₂ emissions, 122 categories are allocated through 1-to-1 mappings and 25 categories through 1-to-many mappings. If detailed categories supporting 1-to-1 mapping are not reported, allocations may rely more on 1-to-many mappings at more aggregated (parent) levels.

17. Given the broad country coverage and long time series considered, country- and year-specific allocation coefficients are essential. While the 2018 OECD methodology derived these coefficients according to the output share of each industry involved in the split (Flachenecker, Guidetti and Pionnier, 2018^[11]), the proposed methodology extends this approach by considering additional variables from a wider range of auxiliary datasets to improve the plausibility of industry allocations. The specific selection differs across the various emission categories.

18. Allocation coefficients are derived using a hierarchical set of auxiliary data sources, applied consistently across countries. For each CRT category, the preferred auxiliary data source available is used

across the full time series in order to avoid breakpoints in the resulting AEA estimates. Where the preferred auxiliary data source is only available for a limited number of years, it is extended to missing years using the approaches described in Section 3 (e.g. extrapolation or backcasting of PEFA using Energy Balances). Alternative auxiliary data sources from lower levels of the hierarchy are only used where the preferred source is entirely unavailable for a country. The hierarchy of auxiliary data sources applied to different CRT categories is described in the subsections below.

19. It is important to note that, based on the national inventory data for 2021, 29.4% of total CO₂ emissions excluding LULUCF in Annex I countries combined⁶ fall into categories with “1-to-1” mapping using this methodology’s correspondence table. In addition, inventory category 1.A.1.a “Public electricity and heat production” - a special case of “1-to-many” mapping that is in most cases allocated largely or entirely to ISIC D “Electricity, gas, steam and air conditioning supply” - accounts for 30.8% of total CO₂ emissions. Inventory category 1.A.3.b “Road transportation”, which is distributed across all ISIC economic activities and households, accounts for a further 23.3%. Together, these three groups represent 83.4% of total CO₂ emissions. Consequently, ensuring accurate 1-to-1 mappings in the correspondence table and robust allocation of these key 1-to-many categories is crucial for the overall accuracy of the resulting AEA estimates.⁷

Hierarchical application of auxiliary data

20. As described in Section 3, the availability of auxiliary data varies across countries, and the preferred data source may be unavailable for some or all years. To ensure in cases of 1-to-many allocations that emissions are allocated using the best available information for each country, a hierarchy of auxiliary data sources is defined for different emission categories. The sub-sections below describe these hierarchies for all emission categories; categories not identified as special cases fall under either fuel combustion (general) or other emission categories (general). The hierarchy is based on conceptual considerations and empirical testing of the performance of the various auxiliary datasets in approaching official AEA estimates as further discussed in Section 5.

CRT 1.A Fuel combustion (general)

21. For CRT categories under 1.A Fuel combustion, except special cases described below such as road transport, allocation coefficients are derived using the following hierarchical set of auxiliary data sources:

1. PEFA use table: emission-relevant use
2. PEFA use table: total use (i.e. sum of transformation use and end use)
3. IEA Energy Balances
 - a. Industry breakdown complemented by output
 - b. Industry breakdown complemented by intermediate consumption
 - c. Industry breakdown complemented by value added

22. The first auxiliary dataset of choice is PEFA, as it records the use of different energy products by economic activities and households. It is therefore the preferred auxiliary data source when a sufficiently detailed ISIC breakdown is available. Among the various PEFA tables, the preferred choice is the emission-relevant use table, but when this is not available, the total use table is used. Total energy use comprises transformation use (e.g. burning coal to generate electricity) and end use (e.g. using gasoline to power a car). Emission-relevant use is a subset of total energy use and refers to the use of energy products that result in GHG and air pollutant emissions through combustion (Eurostat, 2014_[14]). It excludes non-energy use of fuels (e.g. oil used as feedstock for plastics), where carbon is not immediately released to the

atmosphere, and is therefore prioritised over the total use table. PEFA data are compiled and collected annually.

23. When PEFA data are used, only fossil-fuel energy products are considered such as “hard coal” and “natural gas (without bio)”, excluding “motor spirit” and “transport diesel”, because they are considered most relevant to road transport emissions which are treated separately. Energy use in terajoules is multiplied by the corresponding emission factors from the 2006 IPCC Guidelines (IPCC, 2006^[12])⁸ to derive implied emissions, which form the basis for calculating allocation coefficients.

24. In the absence of PEFA, IEA Energy Balances are used. They present an overview of energy flows within an economy based on the territory principle, showing how energy is supplied, transformed and consumed across sectors, and are available for 156 countries, covering the period 1960-2023 for OECD countries (IEA, 2025^[15]). Energy Balances are organised by “flow”, which can often be mapped to a single ISIC economic activity or households. Where necessary, flows are further disaggregated to more detailed ISIC economic activities applying ratios derived from output (preferred), intermediate consumption, or value added data available from the OECD STAN Database for Structural Analysis (e.g. flows such as “Agriculture and forestry” or “Commercial and public service”).

Other emissions (CRT categories beyond 1.A) (General)

25. For CRT categories beyond 1.A Fuel combustion (for which energy data are not relevant), except special cases described below, allocation coefficients are derived using the following hierarchy:

1. Output: sourced from the OECD STAN Database for Structural Analysis
2. Intermediate consumption: sourced from the OECD STAN Database for Structural Analysis
3. Value added: sourced from the OECD STAN Database for Structural Analysis

26. Emissions within AEA scope other than those from CRT category 1.A Fuel combustion arise mainly from CRT categories 1.B “Fugitive Emissions from Fuels”, 2 “Industrial Processes and Product Use (IPPU)”, 3 “Agriculture” and 5 “Waste”. Among these, IPPU accounted for around 80% of such CO₂ emissions in Annex I countries combined in 2021. CRT categories 3 “Agriculture” and 5 “Waste” are treated as 1-to-1 mappings in the reference correspondence table for CO₂ (see Annex A) and therefore do not require auxiliary data for allocation. CRT category 4 LULUCF is excluded from the scope of AEAs under the current SEEA-CF.

27. Output and intermediate consumption are considered as more relevant proxies than value added, particularly for IPPU. Output serves as an indicator of the scale of production, and is more directly observed than intermediate consumption. Intermediate consumption reflects production inputs such as raw materials, which are linked to process emissions, although it also includes other components less directly related to emissions (such as electricity and various services). By contrast, value added is conceptually less directly related to emissions, as it includes components such as wages and operating surplus which do not reflect physical production processes, while output also incorporates these components. Therefore, all three economic variables remain imperfect proxies for emissions-related production activities. As discussed in Section 5, empirical results do not provide a strong basis for establishing a definitive hierarchical preference among these economic variables.

Special cases

CRT 1.A.1.a Public electricity and heat production

28. Inventory category 1.A.1.a “Public electricity and heat production” is one of the largest CO₂ emission sources in many industrialised economies. This category is primarily associated with ISIC D “Electricity, gas, steam and air conditioning supply”. Autoproducers, such as manufacturing industries that

generate electricity/heat for their own use (whether through dedicated generation or combined heat and power processes), are excluded from 1.A.1.a and included in the sectors where they are generated under the IPCC definitions (IPCC, 2006^[16]; IPCC, 2019^[17]).

29. However, where waste management activities generate electricity or heat as a secondary activity (e.g. waste incineration), the Eurostat AEA Manual recommends allocating emissions to ISIC E37–E39 “Sewerage, waste management and remediation activities”, based on supply-table information (Eurostat, 2015^[8]). Evidence from the Netherlands’ detailed data confirms that such emissions may be non-negligible in specific countries (Netherlands Emissieregistratie, 2025^[18]).

30. Allocation follows this hierarchy⁹:

1. PEFA supply table: Supply of electrical energy (P26) and heat (P27)¹⁰
2. National Accounts: Supply and Use Table (SUT), Supply of electricity, gas, steam and air conditioning

If neither source is available, emissions are allocated entirely to ISIC D.

31. In the PEFA supply table, supply of electrical energy (P26) and heat (P27) by ISIC D “Electricity, gas, steam and air conditioning supply” and E37–E39 “Sewerage, waste management and remediation activities” is used to derive allocation coefficients.

32. Where PEFA data are unavailable, the supply table in national accounts is consulted to assess the supply of product “Electricity, gas, steam and air conditioning” by ISIC D and E37–E39. The supply table in national accounts describes the supply of goods and services by product and by type of supply (domestic production or imports) in monetary terms.

CRT 1.A.3.b Road transportation and related categories

33. Road transport emissions are relevant to virtually all economic activities and households, making their allocation particularly complex.

34. In this paper, it is tentatively assumed that emissions by non-residents on the territory and emissions by residents abroad broadly offset one another – an assumption that warrants further investigation and refinement. The magnitude of territory–residence adjustments is highly country-specific and depends on factors such as geographical characteristics (e.g. country size and land borders), degree of economic integration with neighbouring countries, and relative fuel prices, making residence-based estimation particularly challenging. Where available, Tourism Satellite Account data could help assess the magnitude of differences between emissions from non-residents within the territory and those from residents abroad.

35. Emissions reported under 1.A.3.b.iv “Motorcycles” (where separately identified) are allocated to households (1-to-1 mapping).¹¹ Emissions under 1.A.3.b.iii “Heavy-duty trucks and buses” are allocated across industries only. Remaining road transport emissions (e.g. Cars, Light-Duty Trucks) are allocated across both industries and households.

36. CRT 1.A.3.b “Road transportation” and its subcategories are first divided into diesel oil and gasoline using the fuel shares reported in the CRT.¹² Where the selected auxiliary dataset allows separate treatment of these fuels, fuel-specific allocation coefficients are applied to distribute emissions across industries and households.

37. Allocation coefficients are derived using the following hierarchical set of auxiliary data sources:

1. PEFA use table: Emission-relevant use of motor spirit (P14) and transport diesel (without bio) (P17)¹³
2. PEFA use table: Total use of P14 and P17

3. Official SUTs of sufficient granularity: Use of gasoline and diesel
4. Average allocation coefficients derived from European countries' road transport emissions

38. Energy Balances are not suitable for allocating road transport emissions across industries, as they aggregate all road transport fuel use under the single flow "Road", without distinguishing industries and households. Therefore, PEFA is the auxiliary dataset of choice if it has a sufficiently detailed ISIC breakdown. Where PEFA is unavailable, official SUTs are prioritised, provided that the product and industry breakdowns are sufficiently detailed to support allocation (e.g. gasoline and diesel, rather than aggregated petroleum products).¹⁴

39. In the absence of detailed official data on diesel and gasoline use, average allocation coefficients derived from European countries' road transport emissions are used as a fourth-priority option.

40. The resulting industry/households emissions distribution of 1.A.3.b Road transportation is then applied to 2.D Non-energy products from fuels and solvent use, given its link to transport-related products (e.g. transport lubricant). This assumption is unlikely to materially affect overall estimation results, as category 2.D accounts for less than 1% of total CO₂ emissions based on Annex I countries' national inventories data. Emissions from 1.A.2.g.vii Off-road vehicles and other machinery are allocated using a common coefficient derived from detailed UK data, reflecting the lack of off-road specific data for most other countries; further refinements may be explored.¹⁵

Air and maritime transport emissions

41. The proposed methodology does not allocate air and maritime transport emissions (including fishing) from national inventory categories to AEAs. Instead, it directly uses OECD experimental estimates of CO₂ emissions from air and maritime transport. Residence-based emissions available in this dataset capture transport activities operated by resident companies, irrespective of where fuel is purchased, including emissions associated with bunkering abroad.

42. At present, the OECD experimental estimates of CO₂ emissions from air and maritime transport do not include an ISIC breakdown. Consequently, emissions from air, maritime transport (all vessels except fishing) and fishing vessels are allocated to ISIC Rev.4 H51 (Air transport), H50 (Water transport) and A03 (Fishing and aquaculture), respectively, in the proposed methodology.¹⁶

43. The OECD experimental estimates of CO₂ emissions from air and maritime transport do not cover military aircraft or vessels. In national GHG inventories, emissions from military mobile sources are reported under category 1.A.5.b "Mobile" within non-specific fuel combustion activities. As a result, military transport emissions are included in the AEA estimates only to the extent that they are reported by countries to the UNFCCC.

3 Data sources

44. Under the “inventory-first” approach, the geographical and temporal coverage of AEAs is largely determined by the availability and structure of emissions inventory data, as well as by the auxiliary data required to derive the annual allocation matrices. This section summarises the data sources used in the updated methodology and explains how data limitations, such as gaps in time series of data sources, are addressed. The focus is on OECD countries that do not currently report official total-economy AEA estimates.

Inventories

45. The quality of inventory data inputs has a direct influence on the resulting AEA estimates under the inventory-first approach. Reliance on unofficial inventory estimates may lead to substantial over- or under-estimation of total and sectoral emissions as they can differ considerably from the emission levels reported in official inventories (Cárdenas Rodríguez et al., 2024^[19]). In contrast to some other international initiatives for estimating AEAs, the proposed methodology prioritises alignment and consistency with official inventory data as the primary emission data source.

46. Unofficial inventory datasets are used only where necessary to complement sectoral breakdowns and extend time series as described below.

Official inventories

47. The current methodology uses the UNFCCC Common Reporting Tables (CRT) as the primary source of GHG emissions inventory data. Under the Enhanced Transparency Framework (ETF) of the Paris Agreement, all Parties to the Paris Agreement (i.e. countries and the European Union) are required to report national GHG inventories using the CRT in accordance with the modalities, procedures and guidelines adopted by the Conference of the Parties in 2018 (UNFCCC, 2018^[20]). GHG inventory data are broken down by emission source categories in line with the IPCC Guidelines as explained in Section 2. The CRT is the standard reporting format for all Parties since 2024 with no distinction between Annex I and non-Annex I countries. All OECD countries have submitted CRTs to the UNFCCC by 1 October 2025. The primary use of CRT therefore plays a crucial role in prioritising official emissions data and expanding the geographical coverage of AEA estimates.

48. Prior to the introduction of the CRT, the Common Reporting Format (CRF) was the standard reporting format, which was used only by the UNFCCC Annex I Parties.¹⁷ The CRT and CRF use very similar structures and emission source categories.¹⁸ Where relevant, the weights across subcategories available from CRF are used to restore the breakdown of CRT categories for the specific country-year combinations.

Unofficial inventories

49. Since 2024, the International Energy Agency (IEA) has disseminated the dataset “Greenhouse gas emissions estimates matching the energy module of 2006 IPCC Guidelines” (IEA, 2025^[21]). This

dataset provides emissions estimates for carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O). Although limited to IPCC Category 1 “Energy”, it offers extensive geographical coverage and a long time series, and its classification follows the IPCC Guidelines. In limited cases, the IEA dataset provides a more detailed breakdown of emission source categories than official CRT submissions. In such cases, weights across subcategories derived from the IEA estimates are used to complement CRT breakdowns while preserving the level of the parent category.

50. Another unofficial inventory dataset consulted is the HISTCR scenario, which prioritises country-reported data, from PRIMAP-hist. It combines several published sources to create a comprehensive dataset of GHG emissions for all Kyoto gases, covering years from 1750 to 2024 and all UNFCCC member states as well as most non-UNFCCC territories (Gütschow, Busch and Pflüger, 2025^[22]). Although its sectoral detailing is limited, PRIMAP-hist is regarded as the unofficial source most closely aligned with national inventory data (Cárdenas Rodríguez et al., 2024^[19]). It is therefore used in this paper as an interim source for extending official inventory time series until more complete inventory emissions estimates become available from the OECD Environment Directorate, as described below.

Extension of time series

51. Table 2 presents the availability of inventory data for OECD countries that did not report total-economy GHG AEAs to the OECD in the 2025 data collection for dissemination purposes. AEA compilation is mandatory in the European Union, and AEAs for EU Member States and other countries participating in the European Statistical System are collected by Eurostat.

Table 2. Availability of inventory data after 1990

Country	CRT	IEA	PRIMAP-hist
Australia	July 1989 - June 2023 (Fiscal year)	July 1989 – June 2023 (Fiscal Year)	1990-2023
Chile	1990-2022	1990-2023	1990-2023
Colombia	1990-2021	1990-2023	1990-2023
Costa Rica	1990-2021	1990-2023	1990-2023
Israel	1996, 2000, 2003-2022	1990-2023	1990-2023
Korea	1990-2022	1990-2023	1990-2023
Mexico	1990-2022	1990-2023	1990-2023
United States	1990-2022	1990-2023	1990-2023

Note: Data as of 1 October 2025

Source: OECD Data Explorer, “Air emissions - Greenhouse gas emissions Inventories” (OECD, 2026^[23]), and UNFCCC Submission of CRT,¹⁹ IEA “Greenhouse gas emissions estimates matching the energy module of 2006 IPCC Guidelines” (IEA, 2026^[24]), PRIMAP-hist (Gütschow et al., 2016^[25]).

52. The OECD Environment Directorate is currently developing a methodology to fill gaps in the emissions inventories (Pizarro and Drogo, Forthcoming^[26]). The methodology presented in this paper, as an interim method, extrapolates the CRT time series to year $t-2$, but does not interpolate or back-cast missing CRT data.

53. In the absence of official CRT data for the most recent year(s), total emissions are extrapolated using the growth rate of total emissions from PRIMAP-hist. Subcomponents are extrapolated using a contribution-based approach. Specifically, the absolute change in each subcomponent observed in PRIMAP-hist between the two most recent years is translated into a corresponding contribution to the change in total emissions and applied to the latest available official CRT level.

54. Where PRIMAP-hist data are available only at a more aggregated level than the CRT classification, detailed CRT subcategories are extrapolated using the derived growth rate of the corresponding parent category. This assumes similar short-term dynamics within each parent category and, given the close alignment of total emissions between CRT and PRIMAP-hist, provides a consistent and coherent short-term extrapolation until official data or inventory estimates from the OECD Environment Directorate become available.

Auxiliary data sources

Physical energy data

Physical Energy Flow Account

55. The PEFA Use Table serves as the primary auxiliary source for the pro-rata allocation of emissions from CRT category 1.A “Fuel Combustion” to ISIC industries and households, when available. The challenge is that PEFA (when they are available for countries) are not available for the whole time series and therefore require extension, as described below.

56. Where available, the PEFA Supply Table is also used to assess electricity and heat supplied by ISIC D “Electricity, gas, steam and air conditioning supply” and E37–E39 “Sewerage, waste management and remediation activities” for the allocation of emissions from CRT 1.A.1.a “Public electricity and heat production”. This approach is discussed below in combination with the national accounts supply tables.

IEA Energy Balances

57. IEA Energy Balances are used in two main ways: (i) to derive allocation coefficients for fuel-combustion emissions when PEFA are unavailable for all years, and (ii) to extend PEFA time series where they are available only for limited years.

58. As explained in Section 2, “flows” in Energy Balances are mapped to ISIC economic activities and households to derive industry/households allocation coefficients.

Extension of time series

59. Table 3 presents the availability of energy data for the countries covered in Table 2. These data sources are used to allocate emissions from fuel combustion (general). Although longer PEFA time series may exist in national dissemination,²⁰ they are not used here because they are often reported using regional industry classifications that are not fully aligned with ISIC. When data are available for only a single year, the observed year is treated as the reference year for the entire time-series.

Table 3. Energy use data availability at the OECD/IEA Data Explorer after 1990

Country	1 – PEFA emission-relevant use table	2 – PEFA (Total) use table	3 – Energy Balances
Australia	NA	July 2019 – June 2024 (Fiscal year) (aggregated)	July 1989 - June 2023 (Fiscal year)
Chile	NA	NA	1990-2023
Colombia	2023 (provisional)	2023 (provisional)	1990-2023
Costa Rica	NA	2017-2021	1990-2023
Israel	NA	NA	1990-2023
Korea	NA	NA	1990-2023
Mexico	NA	NA	1990-2023
United States	NA	NA	1990-2023

Note: Due to the lack of ISIC breakdown, Australia's PEFA is currently not incorporated into the AEA calculation. Data sources used to derive allocation percentages for Fuel Combustion (general) are highlighted in **bold** for each country. Data as of 06 February 2026.

Source: OECD Data Explorer "Physical Energy Flow Accounts: emissions-relevant use" (OECD, 2026^[27]), "Physical Energy Flow Accounts: use by activity" (OECD, 2026^[28]), IEA Data Explorer "World extended energy balances" (IEA, 2026^[29]).

60. Energy use data from PEFA are first converted to implied emissions using the IPCC emission factors as explained in Section 2. Back-recalculation (Back-casting) of PEFA for the initial years of the time series (i.e. from the first available year to 1990) and extrapolation for the last years of the time series (i.e. from the last available year to $t-2$ (currently 2023)) are performed for each energy product-activity pair. Annual growth rates are derived from the corresponding combinations of energy products and flows in the IEA Energy Balances and applied at this detailed level.²¹

61. For PEFA energy products "Motor spirit (without bio)" and "Transport diesel (without bio)", back-recalculation and extrapolation are based on the IEA products "Motor gasoline excluding any biofuel component" and "Gas/diesel oil excl. biofuels", combined with the IEA flow "Road", irrespective of specific economic or household activity. Consequently, a uniform growth rate is applied across industries and households when extending the use of these fuels, preserving the industry/household shares observed in the first or last available year. This extrapolation approach could be refined using country-specific data instead of the Energy Balances, if available (e.g. vehicle kilometres by industry and households).²²

62. Back-recalculation of PEFA is relevant not only for Costa Rica, as presented in Table 3, but also for estimating AEAs for European countries. In the European Union, AEAs have been mandatory since 2008 and PEFA since 2014 (European Union, 2011^[5]).

63. In case there is a relatively short overlap between official PEFA and IEA Energy Balances compared to the extended time series, it is important to exercise caution when interpreting results.

Economic variables

OECD STAN Database

64. Economic variables such as intermediate consumption, output, and value added are drawn from the OECD STAN Database for Structural Analysis, which provides long and relatively consistent time series across countries (OECD, 2025^[30]).

65. By integrating national accounts with detailed secondary sources – such as structural business statistics, industrial surveys, supply–use tables, and earlier vintages of national accounts – STAN provides harmonised and consistent industry detail under ISIC Rev.4, while ensuring alignment with national accounts totals (Horvát and Webb, 2020^[31]). This supports the granular industry breakdown of the resulting AEA estimates.

66. Table 4 shows the availability of economic variables for the same geographical coverage as Table 2. These data are used to allocate emissions from CRT categories beyond Fuel Combustion (general).

Table 4. Data availability of STAN Database for Structural analysis after 1990

Country	1 – Output	2 – Intermediate Consumption	3 – Value added
Australia	NA	NA	July 1989 – June 2024 (Fiscal year)
Chile	1995-2023	1995-2023	1990-2023
Colombia	1995-2022	1995-2022	1990-2023
Costa Rica	1991-2023	1991-2023	1991-2024
Israel	NA	NA	1995-2024
Korea	1990-2022	1990-2022	1990-2023
Mexico	1990-2023	1990-2023	1990-2023
United States	1990-2023	1990-2023	1990-2023

Note: Data sources used to derive allocation percentages for emissions beyond Fuel Combustion (general) are highlighted in **bold** for each country. Data as of 06 February 2026.

Source: OECD Data Explorer “STAN Database for Structural Analysis, 2025 edition” (OECD, 2025^[30])

67. Where gaps occur in STAN variables or subcategories, series are back-calculated or extrapolated using growth rates of parent categories. Where intermediate consumption and output are entirely missing for certain years, value-added growth rates are used to ensure consistent temporal coverage of the auxiliary dataset over the full estimation period.

Supply and Use Tables (SUT), supply of electricity and heat

68. Table 5 presents the availability of data used to derive allocation coefficients for CRT 1.A.1.a “Public electricity and heat production” between ISIC D “Electricity, gas, steam and air conditioning supply” and E37-E39 “Sewerage, waste management and remediation activities”. In most countries, electricity and heat supplied by ISIC E37–E39 are zero or negligible compared to ISIC D. Consequently, no time-series extension is needed. Where data are unavailable for specific years, the country-specific average allocation coefficients between ISIC D and E37–E39, calculated from years with available data, are applied to the missing years.

Table 5. Data availability on supply tables

Country	1 – PEFA Supply Table	2 - National Accounts, Supply table
Australia	July 2019 – June 2024 (Fiscal year) (aggregated)	July 2010 – June 2023 (Fiscal year)
Chile	NA	2018-2022
Colombia	2023 (provisional)	2014-2023
Costa Rica	2017-2021	2012-2021
Israel	NA	2014
Korea	NA	NA
Mexico	NA	2008, 2013, 2018
United States	NA	1997-2021

Note: Due to the lack of ISIC breakdown, Australia’s PEFA is currently not incorporated into the AEA calculation. Data sources used to derive allocation percentages for electricity and heat generation are highlighted in **bold** for each country. Data as of 06 February 2026.

Source: OECD Data Explorer “Physical Energy Flow Accounts: supply by activity” (OECD, 2026^[32]), “SUT Supply” (OECD, 2026^[33])

OECD estimates on CO₂ emissions from air and maritime transport

69. OECD experimental estimates of CO₂ emissions from air and maritime transport provide timely, residence-based data that support the temporal coverage of AEA estimates (OECD, 2025^[34]; OECD, 2026^[35]). These estimates are based on Automatic Dependent Surveillance–Broadcast (ADS-B) flight data for aviation and Automatic Identification System (AIS) data for maritime transport, ensuring a consistent treatment of flights and voyages across countries (Clarke et al., 2022^[36]; Clarke et al., 2023^[37]).

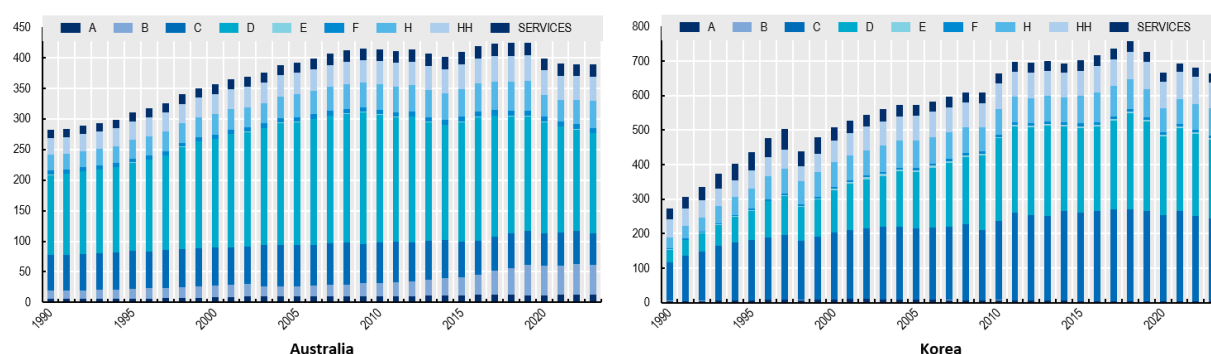
70. For years prior to the availability of these datasets (2013 for aviation and 2019 for maritime), residence-based air and maritime transport emissions are back-calculated using growth rates derived from the corresponding inventory categories – namely 1.A.3.a “Domestic aviation” and the international aviation memo item for air transport, 1.A.3.d “Domestic navigation” and the international navigation memo item for maritime transport, and 1.A.4.c.iii “Fishing” for fishing vessel emissions.

4 Preliminary results

71. Figure 1 presents AEA estimates, excluding LULUCF, for the improved temporal coverage from 1990 to year $t-2$ (2023) broken down by aggregate industries and households for two selected countries, i.e. Australia and Korea, for illustrative purposes. Other country examples are analysed in more detail in the next section through comparisons with official/historical AEAs. Historical AEAs refer to AEA estimates previously compiled and disseminated by countries, for example through one-off pilot estimates available on national dissemination platforms or through data reported to the OECD via the AEA questionnaire prior to 2025. Their availability is summarised later in Table 6. Unless otherwise indicated (e.g. official or historical AEAs), the AEA estimates presented in this paper are produced by the OECD for analytical purposes.

Figure 1. CO₂ emissions with industry/household breakdown

Unit: Million tonnes



Note: ISIC A “Agriculture, forestry and fishing”, B “Mining and quarrying”, C “Manufacturing”, D “Electricity, gas, steam and air conditioning supply”, E “Water supply; sewerage, waste management and remediation activities”, F “Construction”, and H “Transportation and storage”. HH stands for households. Australia’s fiscal year is labelled with the end-year (i.e. fiscal year 1990 refers to the period from June 1989 to July 1990).

Source: OECD calculations

72. Australia’s CO₂ emissions rose steadily from 1990 to 2009, stabilised during 2010–2014, and increased again from 2015 before declining sharply in 2020 due to the COVID-19 crisis. Emissions have since remained below pre-pandemic levels. Electricity and heat supply (ISIC D) is the dominant source of emissions throughout the period, although its contribution declines after 2010. Emissions from Transport (ISIC H) dropped in 2020 due to the pandemic, but resumed increasing in 2022. The share of mining (ISIC B) increases markedly over time, particularly during the 2010s, and since 2019 it has been the third-largest contributor after ISIC D and manufacturing (ISIC C).

73. Korea’s CO₂ emissions increased rapidly from 1990 through the 2000s, with the exception of the late 90s, probably due to the Asian financial crisis. This rapid increase was driven primarily by strong growth in manufacturing (ISIC C) and electricity and heat generation (ISIC D), which together account for the majority of emissions. Emissions continued to rise into the late 2010s, peaking in 2018, before declining

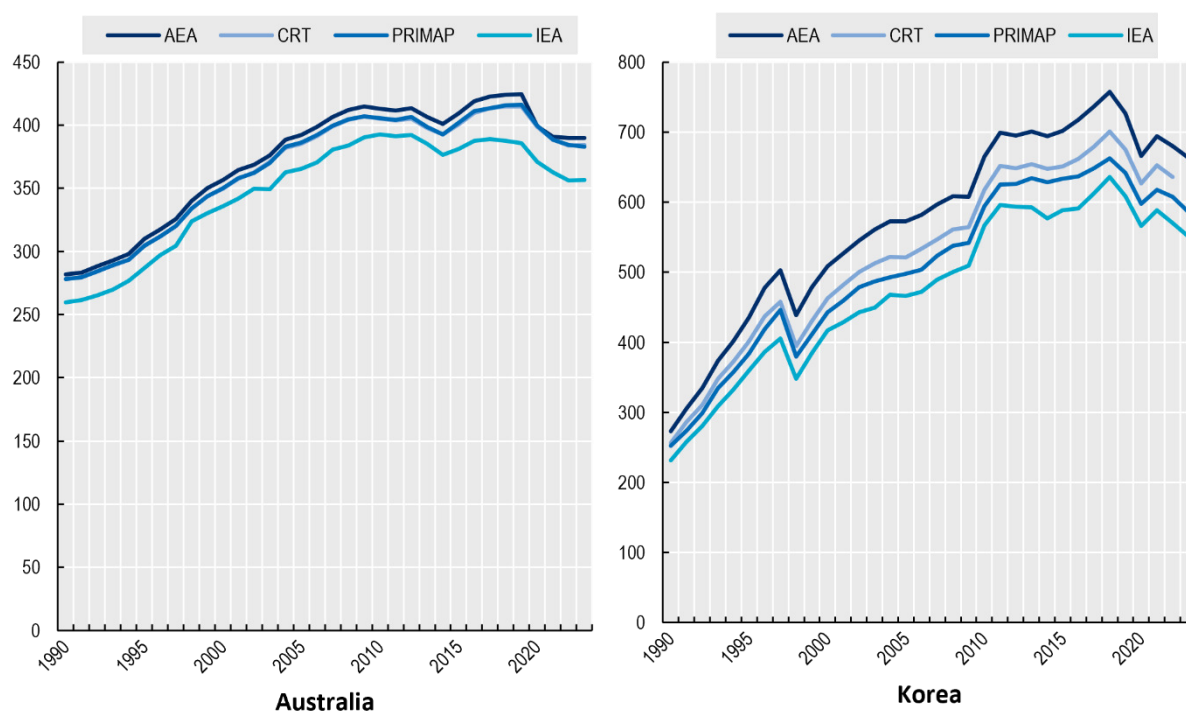
in 2020. Transport (ISIC H) and households (HH) also show steady long-term increases, while other industries contribute relatively smaller shares.

74. Figure 2 compares total emissions from the AEA estimates with national inventory totals (CRT and PRIMAP), as well as with energy-related emissions in the IEA database, for the two countries shown in Figure 1. Emission levels from CRT and PRIMAP are often very similar, especially for UNFCCC Annex I countries (Cárdenas Rodríguez et al., 2024^[19]), with near overlap in some cases (e.g. Australia). IEA estimates are generally lower, reflecting their coverage of energy-related emissions only.

75. Overall, the trends and fluctuations in the AEA estimates are well aligned with those observed in the other emissions datasets. AEA estimates are typically higher than inventory totals because they include international aviation and maritime transport, which are excluded from national inventory totals.

Figure 2. Total CO₂ emissions: AEA, CRT/PRIMAP and IEA

Unit: Million tonnes



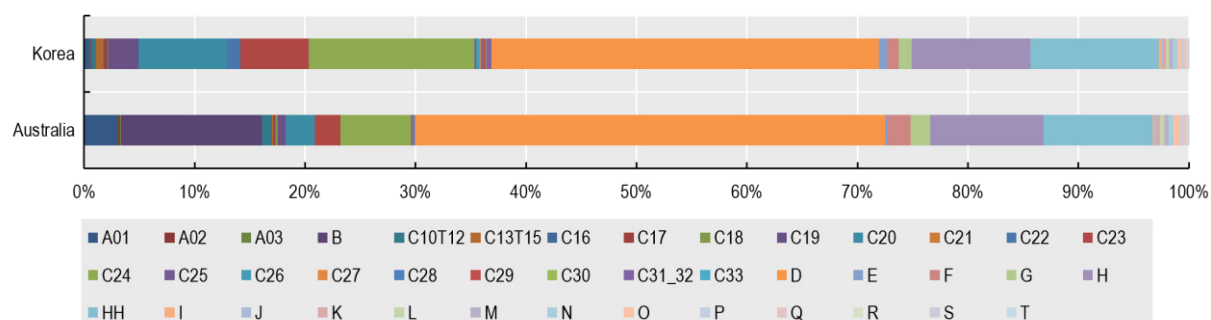
Note: Total emissions in CRT and PRIMAP exclude LULUCF. Australia's fiscal year is labelled with the end-year for AEA, CRT and IEA (i.e. fiscal year 1990 refers to the period from June 1989 to July 1990). The PRIMAP dataset description paper does not specify whether data for Australia are estimated on a calendar-year or fiscal-year basis (Gütschow et al., 2016^[25]).

Source: OECD calculations for AEA, (OECD, 2026^[23]) for CRT, (Gütschow, Busch and Pflüger, 2025^[22]) for PRIMAP, (IEA, 2026^[24]) for IEA.

76. Figure 3 presents industry and household shares of CO₂ AEA estimates for the same two countries with a more detailed industry breakdown ("detailed" breakdown as in Table 7 in the next Section). Comparing the two countries, ISIC A01 "Crop and animal production, hunting and related service activities" and B "Mining and quarrying" account for larger shares in Australia, while C23 "Manufacture of other non-metallic mineral products" and C24 "Manufacture of basic metals" account for larger shares in Korea.

A further breakdown at the A*64 industry level is possible²³, however, results at that level are not shown due to space constraints.

Figure 3. Industry and household CO₂ emissions share in 2022



Note: A01: Crop and animal production, hunting and related service activities, A02: Forestry and logging, A03: Fishing and aquaculture, B: Mining and quarrying, C10T12: Manufacture of food products; beverages and tobacco products, C13T15: Manufacture of textiles, wearing apparel, leather and related products, C16: Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials, C17: Manufacture of paper and paper products, C18: Printing and reproduction of recorded media, C19: Manufacture of coke and refined petroleum products, C20: Manufacture of chemicals and chemical products, C21: Manufacture of basic pharmaceutical products and pharmaceutical preparations, C22: Manufacture of rubber and plastics products, C23: Manufacture of other non-metallic mineral products, C24: Manufacture of basic metals, C25: Manufacture of fabricated metal products, except machinery and equipment, C26: Manufacture of computer, electronic and optical products, C27: Manufacture of electrical equipment, C28: Manufacture of machinery and equipment n.e.c., C29: Manufacture of motor vehicles, trailers and semi-trailers, C30: Manufacture of other transport equipment, C31_32: Manufacture of furniture; other manufacturing, C33: Repair and installation of machinery and equipment, D: Electricity, gas, steam and air conditioning supply, E: Water supply; sewerage, waste management and remediation activities, F: Construction, G: Wholesale and retail trade; repair of motor vehicles and motorcycles, H: Transportation and storage, HH: Activities by households, I: Accommodation and food service activities, J: Information and communication, K: Financial and insurance activities, L: Real estate activities, M: Professional, scientific and technical activities, N: Administrative and support service activities, O: Public administration and defence; compulsory social security, P: Education, Q: Human health and social work activities, R: Arts, entertainment and recreation, S: Other service activities, T: Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use

Source: OECD calculations

5 Discussion and plausibility checks

Key improvements

77. Applying the methodology and data sources described in Sections 2 and 3 addresses the limitations of the previous methodology, particularly regarding industry detail, geographical and temporal coverage, and alignment with SEEA emission boundaries and residence principle.

78. First, the proposed methodology delivers AEA estimates at the A*64 industry level plus households, consistent with the detailed templates of the global AEA questionnaire and the Eurostat AEA questionnaire, thereby enhancing international comparability and policy relevance. The quality of these estimates is assessed in the next subsection.

79. Second, by using the CRT as the primary inventory emission data source and a broader range of auxiliary data, the presented methodology extends geographical coverage and can now be applied to all OECD member countries, including those that did not submit total-economy GHG AEAs for dissemination purposes in 2025, i.e. Australia, Chile, Colombia, Costa Rica, Israel, Korea, Mexico, and the United States. Moreover, the use of the CRT and a wide range of auxiliary data, combined with time-series extension techniques, expands temporal coverage. The methodology enables the estimation of AEAs from 1990 (a typical starting year for national inventory compilation) up to year $t-2$ (2023 as of the pilot estimation).

80. Finally, the proposed methodology aligns closely with the emission boundaries of the SEEA Central Framework (SEEA-CF) by including international transport, while excluding Land Use, Land-Use Change and Forestry (LULUCF) and CO₂ emissions from biomass. The exclusion of LULUCF is in line with the AEA boundary in the SEEA-CF 2012, (Eurostat, 2015^[8]; United Nations, 2014^[13]).²⁴ CO₂ emissions from biomass are distinguished from fossil-fuel emissions, in line with SEEA-CF recommendations, and are excluded from the estimates presented in this paper. International transport is included using OECD experimental estimates of CO₂ emissions from air and maritime transport, allowing the derivation of residence-based emissions for both domestic and international transport.

Plausibility checks: Approach

Country groups

81. The plausibility of the AEA estimates produced by this methodology is assessed by comparing the estimates with existing official AEAs. The benchmarking strategy varies by country group, depending on the availability and scope of official AEAs and auxiliary data. Three country groups are considered.

82. The first group comprises countries that compile official AEAs for the total economy, as well as AEA bridging items, AEAs for road transport emissions, and PEFA. Excluding countries with confidential or a shorter temporal coverage of PEFA/AEA data, or that report bridging items in a different structure, this group comprises 18 European countries: Austria, Czechia, Estonia, Finland, France, Germany, Hungary, Iceland, Ireland, Italy, Luxembourg, Netherlands, Norway, Poland, Portugal, Slovakia, Slovenia, and Spain.

83. For these countries, AEAs excluding emissions from air transport, maritime transport and fishing, and road transport can be constructed using bridging items. AEA estimates derived under the proposed methodology—excluding CRT 1.A.3.b “Road transportation” and OECD experimental air and maritime transport estimates—are benchmarked against official AEAs within this restricted scope. In addition, emissions allocated from CRT 1.A.3.b are compared directly with official AEAs for road transport. This detailed benchmarking allows discrepancies to be analysed by emission source. The availability of auxiliary data, including PEFA, also enables sensitivity analysis to assess the impact of using PEFA, Energy Balances, or monetary variables as allocation bases. In line with the mandatory compilation of PEFA in the EU, analysis for this group focuses on the period from 2014 onwards.

84. The second group includes OECD countries with partial or historical CO₂ AEAs, such as Korea and the United States (historical compilations), and Colombia and Costa Rica (fuel combustion only). For these countries, the temporal coverage of the analysis is aligned with the years for which official AEAs are available (see Table 6). Where necessary, benchmarking is restricted to fuel combustion emissions to ensure comparability in scope.

85. The third group consists of OECD countries that have not yet officially disseminated AEAs for CO₂. For Australia, which has historical AEAs for total GHG emissions, estimated CO₂ AEAs are compared with official GHG totals, where meaningful. For the remaining countries in this group (i.e. Chile, Israel, and Mexico), no AEAs for GHGs have been published. Therefore, the possibilities for quantitative comparisons of estimated emissions by industry are limited.

86. The remaining OECD countries are not considered in the plausibility checks. These are mainly countries that regularly compile total-economy AEAs but do not compile road-transport AEAs or PEFA. Therefore, a separate analysis of total emissions excluding transport is not possible, and the scope for sensitivity analysis using different auxiliary data sources, such as PEFA, is limited. Hence, these countries do not provide additional analytical value compared to Country Group 1. Moreover, these countries are not the primary target group for the application of the proposed estimation methodology, as they already compile official AEAs.

87. Comparison with historical AEAs should be interpreted with care, because the underlying inventory emissions data or other data sources may have been revised in the meantime, or the historical AEAs may have been developed as pilot accounts. AEAs from Australia and Korea were compiled in 2019 and 2018, respectively, while AEAs from United States were reported in 2024.

Table 6. Availability of official CO₂ AEAs

Country	CO ₂ reported	Year	Scope
Australia	Only GHG-total	July 2004 – June 2017 (Fiscal year)	Total-economy
Chile	No	Not applicable	Not applicable
Colombia	Yes	2010-2023(p)	Fuel combustion
Costa Rica	Yes	2017-2021	Fuel combustion
Israel	No	Not applicable	Not applicable
Korea	Yes	2005-2015	Total-economy
Mexico	No (only non-GHG air pollutants)	Not applicable	Not applicable
United States	Yes	2012-2022	Total-economy

Note: Table covers OECD countries that did not report total-economy CO₂ AEAs to Eurostat or the OECD for dissemination purposes in the 2025 data collection. Colombia's 2023 data are provisional.

Source: OECD Data Explorer (OECD, 2026^[38])

Metrics

88. The plausibility of the estimated AEAs is assessed using a set of complementary metrics that jointly evaluate magnitude plausibility, structural misallocation, and growth dynamics. Together, these indicators capture distinct but complementary dimensions of plausibility, enabling a comprehensive assessment of total emission levels, their allocation across economic activities and households, as well as their temporal evolution. These metrics can be computed only when official CO₂ AEAs are available (i.e. not applicable to Country Group 3).

89. Magnitude plausibility at the national level is assessed using the Mean Absolute Percentage Error (MAPE) of total emissions. MAPE measures the average relative deviation between estimated and official total emissions over time and is defined, for each country c , as:

$$MAPE_c = \frac{1}{N_c} \sum_t \left| \frac{\hat{E}_{c,t}^{tot} - E_{c,t}^{tot}}{E_{c,t}^{tot}} \right|$$

where $\hat{E}_{c,t}^{tot}$ and $E_{c,t}^{tot}$ denote estimated and official total emissions for country c in year t , and N_c is the number of years available for comparison.

90. MAPE provides a scale-independent measure of total-level plausibility that is comparable across countries of different size. MAPE focuses exclusively on the accuracy of national totals and does not capture potential misallocation across industries and households.

91. To quantify discrepancies in the industry and household allocation of emissions, the analysis computes the share of misallocated emissions, which measures the minimum proportion of total emissions that would need to be reallocated across industries and households to reproduce the official AEA distribution. For each country-year, total misallocation is calculated as:

$$Misallocation\ share_{c,t} = \frac{1}{2} \frac{\sum_i |\hat{E}_{c,t,i} - E_{c,t,i}|}{E_{c,t}^{tot}}$$

where $E_{c,t,i}$ and $\hat{E}_{c,t,i}$ denote official and estimated emissions in industry (or household) i for country c and year t , and $E_{c,t}^{tot}$ denotes total official emissions. The factor $\frac{1}{2}$ avoids double-counting of over- and under-allocations across industries and households.

92. The misallocation share is expressed as a proportion of total emissions, making it scale-invariant with respect to total emission levels. However, its value depends on the level of industry disaggregation: as the industry classification becomes more detailed, measured misallocation generally increases because over- and under-allocations that offset each other at aggregated levels become visible.

93. This document assesses misallocation at three levels of industry detail in Table 7. The medium level corresponds to the breakdown used in the misallocation analysis in the 2018 OECD methodology (Flachenecker, Guidetti and Pionnier, 2018^[1])

94. In addition to total misallocation, the analysis examines the industry distribution of misallocated emissions, indicating over- or under-allocation in specific industries and households, expressed as a share of total official emissions. This helps identify which industries and households contribute most to overall misallocation and supports diagnostic assessment of the allocation methodology.

Table 7. Industry breakdown for misallocation assessment

ISIC/Households	Aggregate	Medium	Detailed
A: Agriculture, forestry and fishing	X	X	-
A01: Crop and animal production, hunting and related service activities	-	-	X
A02: Forestry and logging	-	-	X
A03: Fishing and aquaculture	-	-	X
B: Mining and quarrying	X	X	X
C: Manufacturing	X	-	-
C10T12: Manufacture of food products; beverages and tobacco products	-	X	X
C13T15: Manufacture of textiles, wearing apparel, leather and related products	-	Combined as other manufacturing	X
C16T18	-	X	-
C16: Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials	-	-	X
C17: Manufacture of paper and paper products	-	-	X
C18: Printing and reproduction of recorded media	-	-	X
C19: Manufacture of coke and refined petroleum products	-	X	X
C20_21	-	X	-
C20: Manufacture of chemicals and chemical products	-	-	X
C21: Manufacture of basic pharmaceutical products and pharmaceutical preparations	-	-	X
C22_23	-	X	-
C22: Manufacture of rubber and plastics products	-	-	X
C23: Manufacture of other non-metallic mineral products	-	-	X
C24_25	-	X	-
C24: Manufacture of basic metals	-	-	X
C25: Manufacture of fabricated metal products, except machinery and equipment	-	-	X
C26: Manufacture of computer, electronic and optical products	-	Combined as other manufacturing	X
C27: Manufacture of electrical equipment	-		X
C28: Manufacture of machinery and equipment n.e.c.	-		X
C29: Manufacture of motor vehicles, trailers and semi-trailers	-		X
C30: Manufacture of other transport equipment	-		X
C31_32: Manufacture of furniture; other manufacturing	-		X
C33: Repair and installation of machinery and equipment	-		X
D: Electricity, gas, steam and air conditioning supply	X	X	X
E: Water supply; sewerage, waste management and remediation activities	X	X	X
F: Construction	X	X	X
G: Wholesale and retail trade; repair of motor vehicles and motorcycles	Combined as services	Combined as services	X
H: Transportation and storage	X	X	X
I: Accommodation and food service activities	Combined as services	Combined as services	X
J: Information and communication			X
K: Financial and insurance activities			X
L: Real estate activities			X
M: Professional, scientific and technical activities			X
N: Administrative and support service activities			X
O: Public administration and defence; compulsory social security			X
P: Education			X
Q: Human health and social work activities			X
R: Arts, entertainment and recreation			X
S: Other service activities			X

T: Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use			X
Households	X	X	X

95. Finally, the ability of the AEA estimates to capture temporal dynamics is assessed by comparing growth rates of total emissions. Correlation coefficients between the growth rates of estimated and official AEAs are used as the primary metric to assess whether estimated AEAs replicate observed growth patterns. The mean absolute error of growth rates and directional accuracy (i.e. whether the signs of growth rates match) are used as complementary metrics.

96. In addition to total emissions, sectoral growth rates are examined for selected key industries (at the aggregate level) to provide further insight into the consistency of estimated emission trends at a more detailed level, while recognising that sector-level growth rates may be more volatile and sensitive to allocation assumptions.

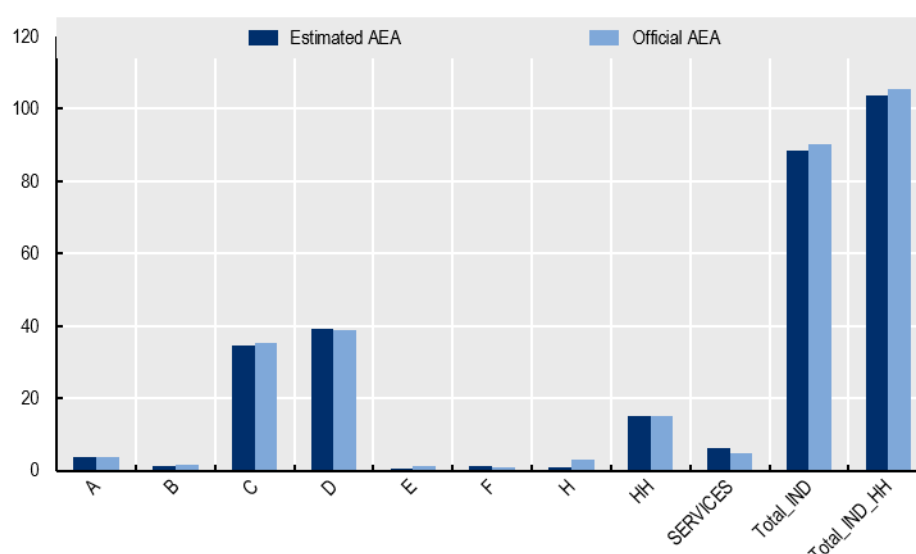
Plausibility checks: Results

Overall results

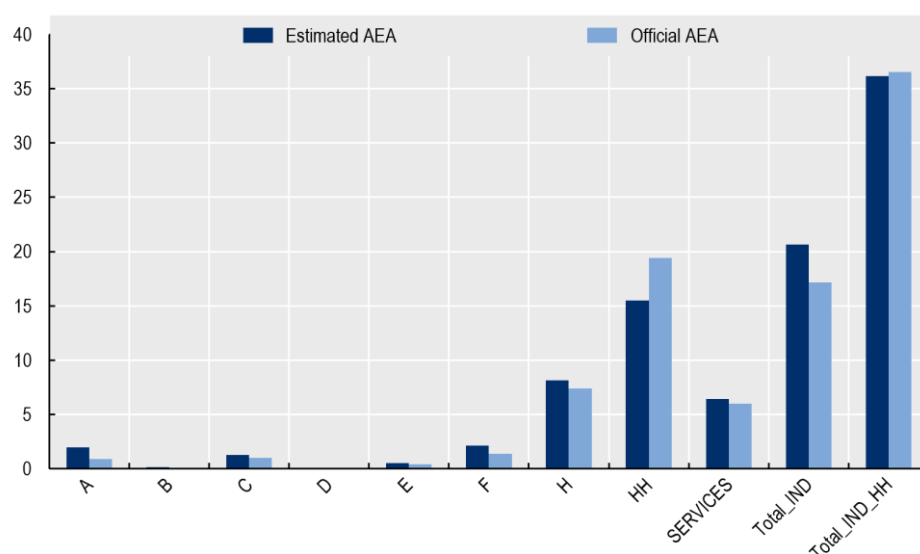
97. Figure 4 compares OECD estimates with official AEAs for Country Group 1 by industries and households. Emissions are presented as simple averages across all countries and years over the period 2014–2023. Overall, OECD AEA estimates are closely aligned with official AEAs for emissions excluding air transport, maritime transport and fishing, and road transport emissions, while road transport emissions from households are underestimated. However, over- and under-estimation across countries and years may partially offset each other. Total discrepancies and industry misallocation are examined in greater detail using the metrics defined above in the following subsections and Annex B and Annex C.

Figure 4. Comparison between OECD AEA estimates and official AEAs, Country Group 1, 2014–2023

Total emissions excluding air transport, maritime/fishing and road transport, Unit: Million tonnes



Road transport emissions, Unit: Million tonnes



Note: Emissions are averaged across all country–year observations for the period 2014–2023 (Country Group 1). The allocation basis for AEA estimates is the preferred auxiliary data source (e.g. PEFA). Emissions excluding air, maritime, and road transport. Fishing emissions are excluded for countries that report 1.A.4.c.iii “Fishing” separately. ISIC A “Agriculture, forestry and fishing”, B “Mining and quarrying”, C “Manufacturing”, D “Electricity, gas, steam and air conditioning supply”, E “Water supply; sewerage, waste management and remediation activities”, F “Construction”, and H “Transportation and storage”. HH stands for households. Total_IND and Total IND_HH stand for total industries and total industries and households, respectively.

Source: OECD calculation

Total emissions

Country Group 1

Emissions excluding air transport, maritime/fishing, and road transport

98. Figure 5 reports the MAPE of estimated total CO₂ emissions relative to official AEAs for Country Group 1, excluding air transport, maritime/fishing, and road transport emissions, over the period 2014–2023. The results do not vary with the choice of auxiliary data used for industry allocation, as the allocation basis affects only the distribution of emissions across industries and households and not the total emissions level.

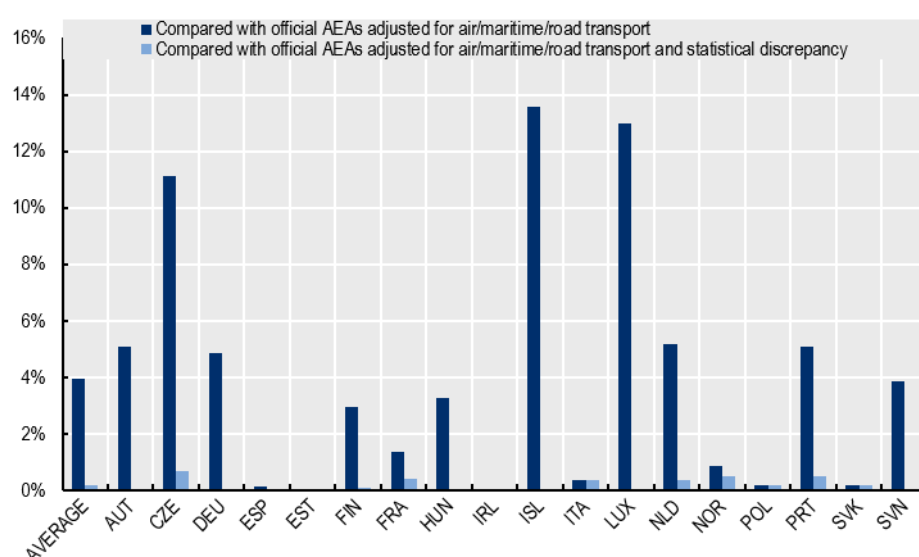
99. Overall, the results show a high degree of consistency. The simple average MAPE across countries (without weighting by country emissions size) is approximately 3.9%. Countries such as Estonia, Ireland, Poland and Slovakia exhibit very low MAPE values, indicating near-perfect reproduction of official totals.

100. In contrast, a few countries display larger discrepancies from official totals (e.g. above 10% for Czechia, Iceland, and Luxembourg). These countries report sizeable bridging items labelled “Adjustments and statistical discrepancy” in total-economy AEAs and/or road transport AEAs to reconcile AEAs with inventories. When official totals are adjusted to include these bridging items, MAPE values decrease substantially for the affected countries, falling to approximately 0.2 % on average. This suggests that a large part of the observed deviation from official totals reflects explicitly reported statistical discrepancies rather than shortcomings of the allocation methodology. Moreover, Luxembourg uses the energy statistics as the first data source and GHG inventories as the second data source to cover emissions from sources beyond energy consumption (Eurostat, 2025_[39]). This may partly explain the discrepancies in total

emissions levels between the official AEAs and the OECD estimates, which are primarily derived from inventories.

101. Overall, the results indicate that the proposed methodology reproduces total CO₂ emissions with high plausibility across countries, for the scope excluding air transport, maritime/fishing, and road transport. For context, the IPCC Guidelines estimates the uncertainty associated with default CO₂ emission factors of fossil fuels at around 5-10% (IPCC, 2006^[16]).²⁵ National inventory reports also typically include uncertainty ranges of a similar magnitude (e.g. $\pm 3.6\%$ for emissions excluding LULUCF in Australia's National Inventory (Australia's Department of Climate Change, Energy, the Environment and Water, 2025^[40])), suggesting that the observed deviations are broadly within the range of reporting uncertainty.

Figure 5. Mean Absolute Percentage Error (MAPE) of total CO₂ emissions (excluding air transport, maritime/fishing and road transport), Country Group 1, 2014-2023



Note: AEAs are estimated based on the preferred auxiliary data sources (i.e. PEFA emission-relevant energy use for Energy and output for non-Energy). Average shows the simple average across countries (without weighting by emissions). Fishing emissions are excluded for countries that report 1.A.4.c.iii "Fishing" separately. AUT: Austria, CZE: Czechia, DEU: Germany, ESP: Spain, EST: Estonia, FIN: Finland, FRA: France, HUN: Hungary, IRL: Ireland, ISL: Iceland, ITA: Italy, LUX: Luxembourg, NLD: Netherlands, NOR: Norway, POL: Poland, PRT: Portugal, SVK: Slovakia, SVN: Slovenia.

Source: OECD calculation

102. The comparison of year-to-year growth rates indicates that estimated AEAs closely track the temporal dynamics of official emissions for total emissions excluding air, maritime, road transport and fishing, with the exception of Iceland. Figure 6 shows that correlation coefficients are generally close to one. In addition, the mean absolute error of growth rates is small (0.6 percentage points, simple average across countries), and the direction of change is correctly identified in most cases (94%). This suggests that the estimation methodology captures both the magnitude and direction of short-term emission changes.

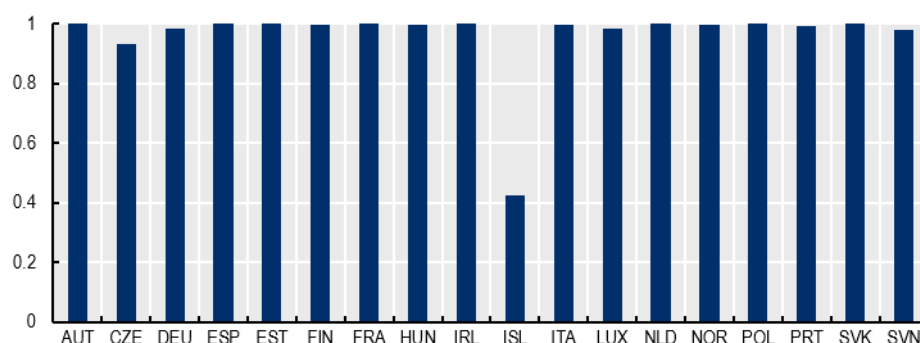
103. At the industry and household level, correlation coefficients are highest for emission-intensive industries such as manufacturing (ISIC C), electricity and gas supply (ISIC D), and households, indicating that the estimates closely reproduce the dynamics of emissions in the most important sources. Lower correlations are observed in sectors accounting for smaller shares of CO₂ emissions, such as mining (ISIC B) and construction (ISIC F), reflecting greater volatility in growth rates. The transport and storage sector (ISIC H) shows lower correlation, given that major transport components (air, maritime and road

transport) are excluded, leaving relatively small and heterogeneous activities (e.g. pipeline transport, rail, warehousing, and postal services).

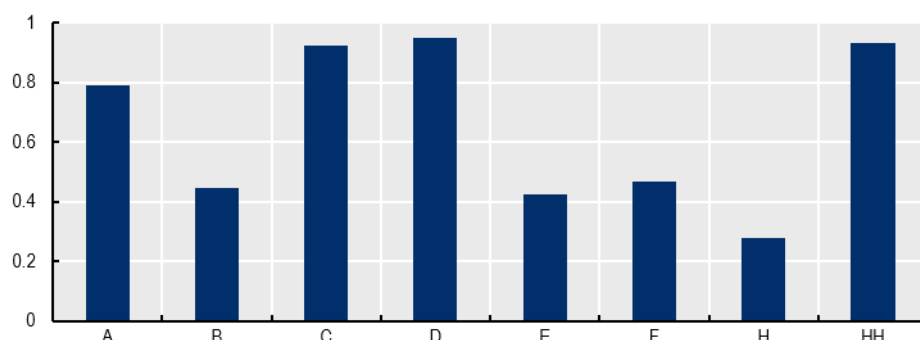
104. Overall, the results suggest that the methodology performs particularly well in capturing the growth rates of total emissions and major emission sources. This supports the use of the methodology detailed in this document to extend the time series of official AEAs where data availability is limited (see Section 6).

Figure 6. Correlation coefficients of growth rates between official and estimated AEAs, 2014-2023

Correlation of growth rates of total emissions by country



Correlation of growth rates of industry/household emissions



Note: For industry and household emissions, correlation coefficients are calculated by pooling observations across all countries and years. ISIC A "Agriculture, forestry and fishing", B "Mining and quarrying", C "Manufacturing", D "Electricity, gas, steam and air conditioning supply", E "Water supply; sewerage, waste management and remediation activities", F "Construction", and H "Transportation and storage". HH stands for households.

Source: OECD Calculations

Road transport emissions

105. Figure 7 presents the Mean Absolute Percentage Error (MAPE) for road transport emissions. The proposed methodology assumes that emissions by non-residents on the territory are offset by emissions from residents fuelling abroad.

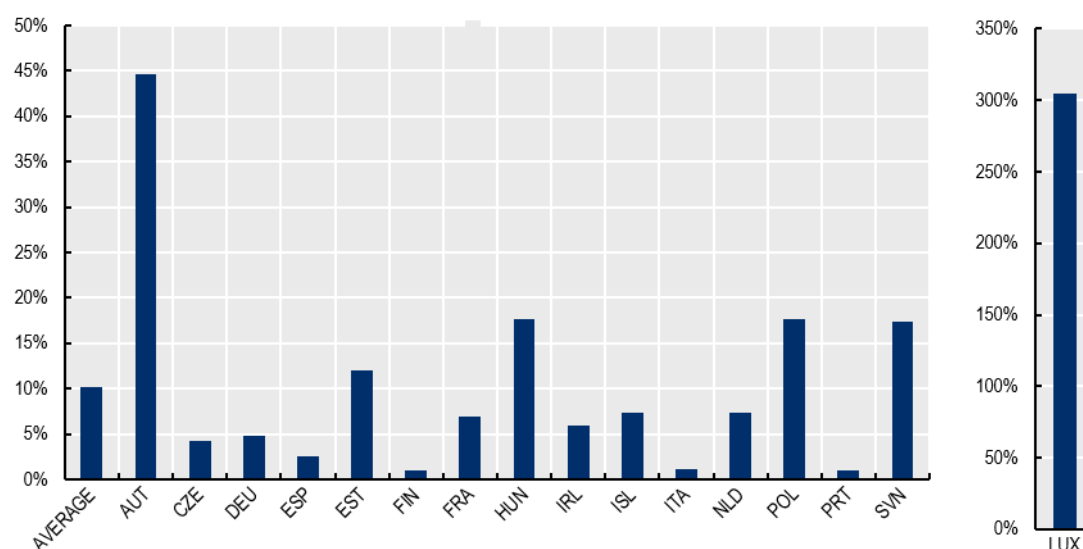
106. MAPE values exhibit considerable variability, suggesting that the simplifying assumption of offsetting non-resident emissions with emissions from residents abroad does not consistently hold across countries. Luxembourg is a clear outlier as a substantial share of road transport emissions on its territory is emitted by non-residents. This reflects the fact that Luxembourg is a transit country and that non-residents from neighbouring countries frequently purchase road transport fuels in Luxembourg because of

its significantly lower fuel prices, a phenomenon referred to as “fuelling tourism” (Eurostat, 2015^[8]). Under the residence principle, emissions associated with fuels purchased by non-residents on the territory are excluded from official AEAs. As a result, the simplifying assumption can lead to particularly high MAPE values for Luxembourg.

107. Countries that are relatively small and/or have extensive land borders—such as Austria, Hungary, Poland, and Slovenia—display larger discrepancies (above 15%), reflecting the sensitivity of road transport emissions to residence–territory adjustments. In contrast, larger countries with more limited land borders and substantial maritime borders—such as Finland, Italy, Portugal, and Spain—show smaller discrepancies (below 3%).

108. These results indicate that the plausibility of road transport emission estimates depends strongly on country-specific circumstances. Among countries in Groups 2 and 3, the simplifying assumption is likely to be more realistic for geographically isolated countries such as Australia and Korea, who do not have freely accessible land borders.

Figure 7. Mean Absolute Percentage Error (MAPE) of road transport emissions, Country Group 1, 2014-2023



Note: AEAs are estimated based on the preferred auxiliary data sources (i.e. PEFA emission-relevant energy use). Norway and Slovakia do not report the bridging item related to land transport or report zero emissions, hence excluded from this Figure. Simple average is calculated excluding Luxembourg. Temporal coverage from 2014 to 2023.

Source: OECD calculations

Country Group 2

109. For Country Group 2, estimated CO₂ AEAs are benchmarked against official CO₂ AEAs in accordance with the reported scope (i.e. total economy or fuel-combustion). Figure 8 presents the Mean Absolute Percentage Error (MAPE) for total emissions per country.

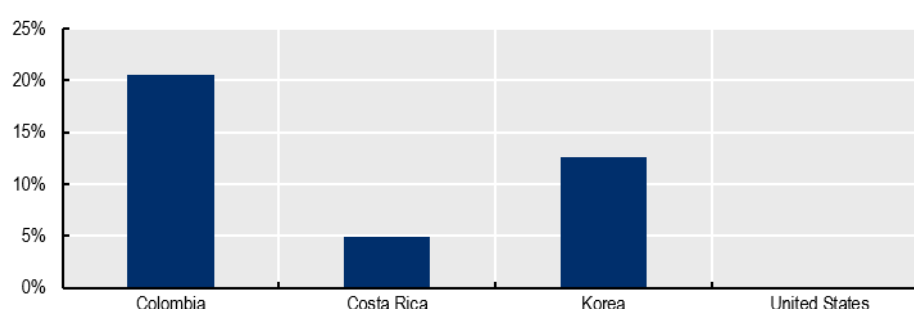
110. The OECD AEA estimates are closely aligned with the official totals reported by the United States in 2024 with a difference of only 0.16%.

111. Larger discrepancies (approximately 12.6%) are observed when comparing OECD estimates with historical official totals reported by Korea. However, ongoing bilateral discussions between Korea and the

OECD indicate that these discrepancies are largely resolved when compared with Korea's pilot AEAs currently under development.

112. Colombia reports air emissions generated from fuel production and consumption processes in its official AEAs. Total CO₂ emissions reported in the official AEAs show substantial discrepancies compared with emissions reported under CRT Category 1 “Energy” in the national inventory, averaging approximately 22.7% higher over the period 2010–2021, while bridging items reconciling the AEAs and the inventories are not reported. Under the OECD methodology, total emissions are anchored to the official inventory total and supplemented with OECD estimates for air and maritime transport. In the absence of official bridging items reconciling the inventory and accounts, it is difficult to analyse the source of these discrepancies in greater detail. Moreover, Colombia applies its PEFA as the primary data source to compile AEAs (DANE, 2025^[41]). This may partly explain the discrepancies in total emissions levels between the official AEAs and the OECD estimates, which are primarily derived from inventories.

Figure 8. Mean Absolute Percentage Error (MAPE) of total CO₂ emissions, Country Group 2, 2014–2023



Note: Colombia and Costa Rica's official AEAs cover only emissions from fuel combustion. Costa Rica's bridging table suggests that only resident emissions for domestic transport are covered for air and maritime transport. OECD estimates are adjusted to align the scope.

Source: OECD calculations

Misallocated emissions

Country Group 1

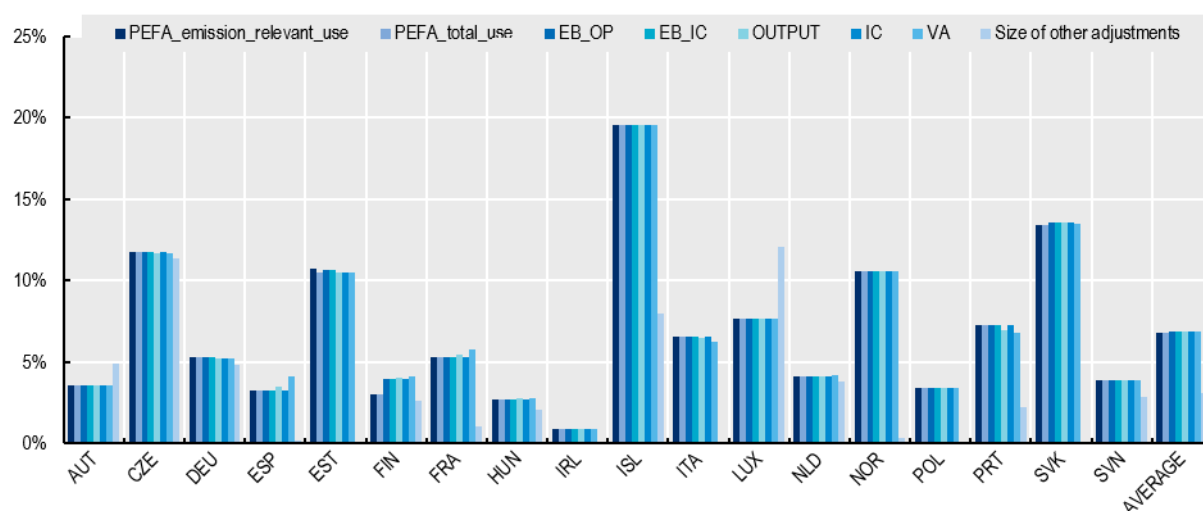
Emissions excluding air transport, maritime/fishing, and road transport

113. Figure 9 presents the share of misallocated emissions, as defined above, for Country Group 1, excluding air transport, maritime/fishing and road transport. The analysis is conducted at the aggregate level of industry breakdown. Results are shown using alternative auxiliary datasets for industry allocation (PEFA (emission-relevant use and total use), Energy Balances, and other economic variables for benchmarking (i.e. intermediate consumption, output, and value added)). The temporal coverage for plausibility checks for Country Group 1 is 2014–2023, thereby avoiding the potential impact of extending PEFA time series using Energy Balances.

114. Countries exhibiting large discrepancies in total emission levels in Figure 5 (e.g. Czechia, Iceland, Luxembourg) also display high misallocation in Figure 9. This reflects the fact that over- or under-estimation at the total level must necessarily be reflected at the industry level, whether distributed across multiple industries (and households) or concentrated in a few. As noted above, part of these discrepancies can be explained by the official bridging item “Adjustments and statistical discrepancy” (e.g. Czechia). By contrast, some countries (e.g. Estonia, Norway, Slovakia) show low total-level deviation in Figure 5 but relatively high misallocation shares in Figure 9, indicating that over- and under-allocation across industries

and households offset each other in aggregate totals. CO₂ emissions from fishing account for a considerable share of total emissions excluding LULUCF in Iceland (approximately 15% on average in 2014-2023). The exclusion of fishing emissions therefore reduces the scope of emissions more markedly for Iceland, which in turn increases the measured misallocation share.

Figure 9. Misallocation share of total CO₂ emissions (excluding air transport, maritime/fishing, and road transport), Country Group 1, 2014-2023



Note: Analysis on the aggregate breakdown level. EB_OP stands for energy balances with industry breakdowns complemented by output. EB_IC stands for energy balances with industry breakdowns complemented by intermediate consumption. VA stands for value added. Temporal coverage from 2014 to 2023. Average shows the simple average across countries (without weighting by emissions). "Other adjustments" refer to emissions reported under the bridging item "Adjustments and statistical discrepancy", expressed as a share of total emissions excluding air, maritime and road transport. Fishing emissions are excluded for countries that report 1.A.4.c.iii "Fishing" separately.

Source: OECD calculation

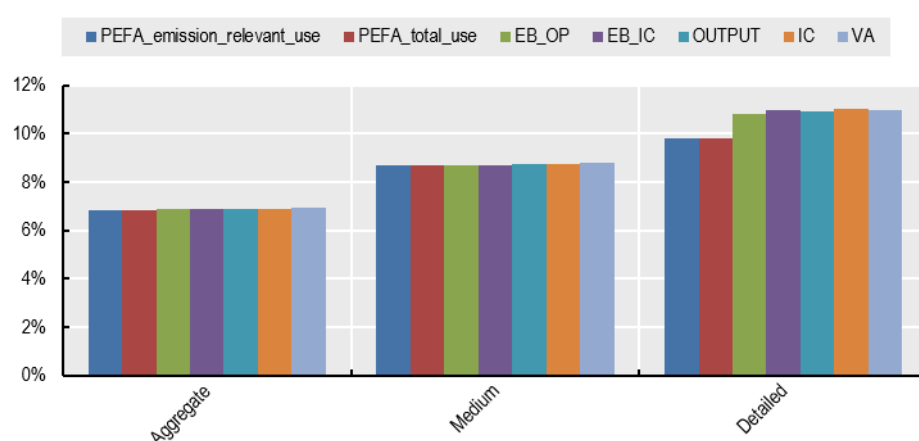
115. Figure 10 further indicates that, at the Aggregate level, the misallocation share is largely insensitive to the choice of auxiliary dataset. As explained above, a substantial portion of emissions falls under 1-to-1 mappings. Moreover, many 1-to-many mappings occur between ISIC divisions within the same section (e.g. A01–A03 within ISIC A "Agriculture, forestry and fishing" or divisions within ISIC C "Manufacturing") or across different service industries. At the aggregate level, these industries are grouped within the same parent section or aggregate service industries, making the allocation relatively insensitive to the specific auxiliary data used.

116. When more detailed ISIC levels are considered, however, the choice of auxiliary data becomes more consequential. As shown in Figure 10, PEFA yields the lowest misallocation shares for the detailed industry breakdown among the alternative allocation bases. The distinction between PEFA emission-relevant energy use and PEFA total use has only a negligible effect on the results.

117. Energy Balances and monetary variables produce broadly similar levels of misallocation at each level of industry breakdown. This partly reflects the fact that the mapping from energy flows to ISIC activities is complemented, where necessary, using economic variables. Comparing Energy Balances complemented by output or intermediate consumption with allocation based solely on the corresponding economic variable, misallocation shares are only marginally lower when Energy Balances are used. Further refinement of the mapping from energy flows to economic activities could improve allocation based on Energy Balances.

118. The choice among output, intermediate consumption, and value added does not materially affect the magnitude of misallocation shares. The impact amounts to approximately 0.04, 0.02, and 0.14 percentage points at the aggregate, medium, and detailed levels, respectively. Moreover, the economic variable associated with the lowest misallocation share varies by level of disaggregation. For example, intermediate consumption yields the smallest misallocation share at the medium level but the largest at the detailed level. These empirical results do not provide a strong basis for establishing a definitive hierarchical preference among the economic variables.

Figure 10. Misallocation share by level of industry breakdown and allocation basis (excluding air transport, maritime/fishing, and road transport), Country Group 1, 2014-2023



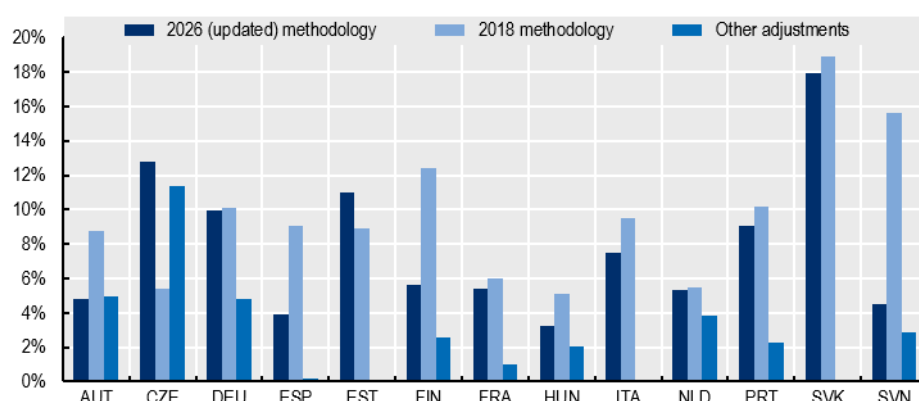
Note: EB_OP stands for energy balances with industry breakdowns complemented by output. EB_IC stands for energy balances with industry breakdowns complemented by intermediate consumption. VA stands for value added. Fishing emissions are excluded for countries that report 1.A.4.c.iii "Fishing" separately.

Source: OECD calculation

119. Figure 11 compares misallocation shares under the 2018 methodology and the updated approach, using the medium industry breakdown and excluding air transport, maritime/fishing, and road transport emissions. A direct comparison is not strictly possible, as the time coverage and the emission scope differ between the two analyses (i.e. the 2018 methodology compared territory-based emissions, including transport, using bridging items). Nevertheless, the updated methodology appears to achieve improved industry allocation in most countries, as shown in Figure 11.

120. As discussed in relation to Figure 5 and Figure 9, part of the discrepancies from official AEAs can be attributed to "adjustments and statistical discrepancies", which are reported as official bridging items for total economy AEAs and road transport AEAs. In some cases (e.g. Austria and Czechia), the misallocation share under the 2026 (updated) methodology is of a similar magnitude to the officially reported statistical discrepancies relative to total emissions within the adjusted scope.

Figure 11. Comparison of misallocation shares under the 2018 and updated methodology, Country Group 1, 2014-2023



Note: Country scope is narrowed down to align with the countries reviewed in the 2018 paper. Misallocation shares are assessed for 2008-2014 under the 2018 methodology and for 2014-2023 under the updated methodology. "Other adjustments" refer to emissions reported under the bridging item "Adjustments and statistical discrepancy" in both total-economy AEAs and road transport AEAs, expressed as a share of total emissions excluding air, maritime and road transport. Fishing emissions are excluded for countries that report 1.A.4.c.iii "Fishing" separately. Source: OECD calculation, (Flachenecker, Guidetti and Pionnier, 2018^[11])

121. The distribution of misallocated emissions across industries and households, excluding air transport, maritime/fishing, and road transport, under the updated methodology is presented in Annex B. The share of misallocated emissions from road transport relative to total road transport emissions, and its distribution, are discussed in Annex C.

Country Group 2

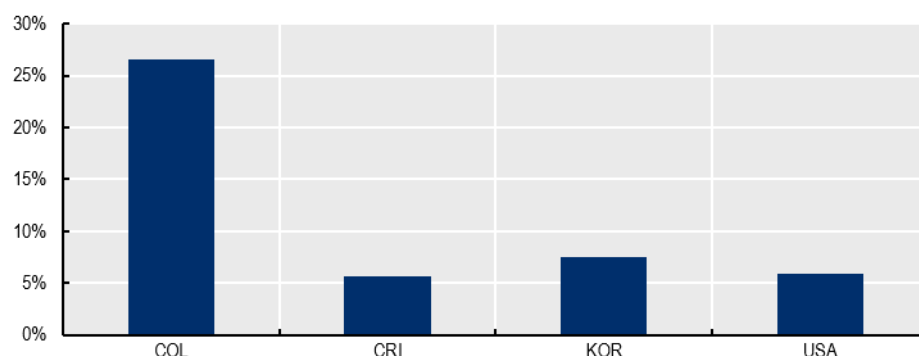
122. Figure 12 presents the share of misallocated emissions for Country Group 2. The average distribution of misallocation is not reported because the sample size is too small. Instead, the paragraphs below describe country-specific industry and household distributions.

123. Consistent with the large discrepancies observed at the total level in Figure 8, Colombia exhibits a high misallocation share. In addition, Colombia's PEFA is currently available only for one year (provisional) on the OECD Data Explorer, which is used as the reference year across the entire time series. Once a longer historical PEFA time series becomes available, the industry allocation may improve. OECD estimates show lower total emissions than the official AEAs, primarily due to underestimation in ISIC C and ISIC D.

124. Comparing OECD estimates with Korea's historical AEAs, emissions appear to be overestimated mainly in ISIC H. However, ongoing bilateral discussions indicate that the OECD estimates align much more closely with Korea's pilot AEAs currently under development.

125. The United States shows a misallocation share of approximately 5.9%, even though OECD estimates are very close to the official AEA totals (Figure 8). This reflects offsetting effects at the industry level: overestimation in ISIC H and underestimation in households cancel out when aggregated. These discrepancies between the transport industry and households may be related to the allocation of road transport emissions, which is based on the European simple average in the absence of PEFA data. The prevalence of private car ownership, more limited public transport coverage, and longer travel distances in the United States could imply that the allocation coefficient of road transport emissions to households should be higher than in Europe.

Figure 12. Misallocation share, Country Group 2, 2014-2023



Note: Colombia and Costa Rica's official AEAs cover only emissions from fuel combustion. Costa Rica's bridging table suggests that only resident emissions for domestic transport are covered for air and maritime transport. OECD estimates are adjusted to align the scope.

Source: OECD calculation

Other checks (Country Group 3)

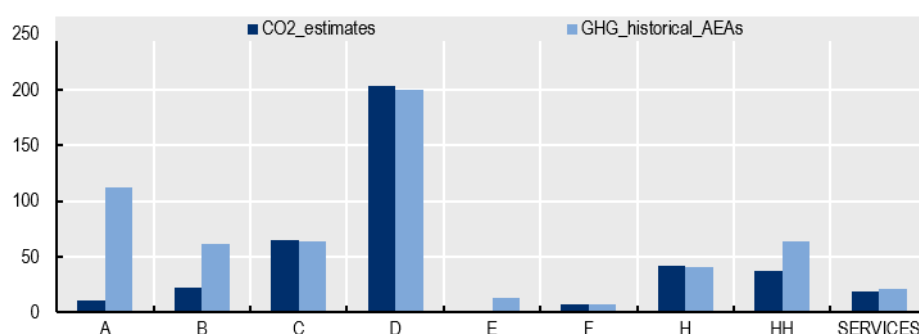
Comparison with official total GHG AEAs

126. Historical AEAs for total GHG are available for Australia. OECD estimates for CO₂ AEAs are therefore compared with historical official GHG AEAs by industry and households.

127. Figure 13 presents industry and household CO₂ emissions in the OECD estimates alongside total GHGs from the historical official AEAs. Industries primarily associated with fuel combustion (e.g. ISIC C, D, and H) exhibit close alignment between CO₂ and total GHG emissions. In contrast, industries associated with methane (CH₄) and nitrous oxide (N₂O) emissions from non-energy sources (e.g. ISIC A, B, and E) show substantially higher total GHG emissions than CO₂ alone. This pattern is consistent with the underlying emission sources and suggests that the industry structure of OECD CO₂ estimates aligns well with the historical GHG AEAs.

Figure 13. Comparison of industry and household emissions between OECD CO₂ estimates and official GHG total of Australia, 2004/05-2016/17

Unit: Million tonnes of CO₂-equivalent



Note: ISIC A "Agriculture, forestry and fishing", B "Mining and quarrying", C "Manufacturing", D "Electricity, gas, steam and air conditioning supply", E "Water supply; sewerage, waste management and remediation activities", F "Construction", and H "Transportation and storage". HH stands for households. Average across fiscal years from 2004/2005 to 2016/2017.

Source: OECD Calculation and OECD Data Explorer (OECD, 2026^[38])

6 Dissemination Strategy

128. The proposed methodology generates total-economy CO₂ AEAs with A*64 industry breakdown for all OECD countries from 1990 to year $t-2$.²⁶ For countries that already disseminate official CO₂ AEAs, this may result in two parallel sets of estimates for the same country–year–industry/household combinations. While this overlap is useful for calibrating and validating the OECD methodology by comparing the two datasets (as discussed in Section 5), dissemination on the OECD Data Explorer should aim to provide users with a single, clear and coherent dataset.

129. The OECD therefore considers dissemination strategies tailored to five country situations:

- i. countries with official AEAs covering a full time series from 1990 to year $t-2$ with A*64 industry breakdown;
- ii. countries with official AEAs covering a limited time series;
- iii. countries with historical AEAs that have been discontinued;
- iv. countries compiling official AEAs for fuel combustion only; and
- v. countries without official AEAs.

In all cases, official AEA data remain unchanged; OECD estimates do not replace official AEAs for the same country–year–industry/household observations. OECD estimates are used only to (a) disaggregate official totals where additional detail is missing, (b) extend time series beyond the period covered by official data, or (c) fill gaps where official data are obsolete or completely missing. All such data points will be clearly flagged, and the data sources used for individual countries will be documented in the metadata.

130. First, for countries for which official CO₂ AEAs with a full time series and A*64 industry breakdown are available and regularly maintained (i.e. excluding historical AEAs), the OECD will disseminate only the official AEAs (currently only applicable to the United Kingdom).

131. Second, for countries with official AEAs covering a limited time series (e.g. mandatory EU reporting starting from reference year 2008), official values should be prioritised for overlapping country–year–industry/household observations, and OECD estimates will not be disseminated where official data are available. Where OECD estimates provide a more detailed industry breakdown, the OECD will use sectoral shares derived from its estimates to disaggregate official totals while preserving the parent-category levels.

132. For backcasting or extrapolating beyond the official reporting period, the OECD will use its estimates to extend the official time series, as they closely replicate the growth patterns of official emissions, particularly for total emissions and emission-intensive industries (see Section 5). The potential use of contribution-based methods to extend official emissions will be explored, following the approach applied to extend the CRT time series described in Section 3.

133. Third, for countries with historical AEAs that have been discontinued several years ago, OECD estimates will be used as the default series for dissemination, reflecting potential revisions to underlying data sources such as national inventories. However, where countries prefer, or where underlying data sources have not undergone substantial revisions, the extension approach described above for regularly compiling countries may instead be applied to preserve historical official values.

134. Fourth, for countries compiling AEAs only for fuel combustion, official AEAs may be backcast or extrapolated using OECD estimates adjusted to the same fuel-combustion scope. OECD estimates for emission sources not covered by the official AEAs (i.e. CRT source categories beyond 1.A “Fuel combustion”) may then be added to the extended fuel-combustion AEAs to derive total AEA estimates.

135. Finally, for countries without official AEAs for CO₂, OECD estimates will be disseminated on the OECD Data Explorer.

7

Next steps

136. This working paper presents an updated methodology for estimating Air Emission Accounts (AEAs) that advances ongoing international efforts to produce robust, internationally comparable, and policy-relevant emissions statistics, which will be an important foundation for applications such as carbon-footprint calculations (Yamano, Lioussis and Cimper, 2024^[2]). Building on the OECD's 2018 methodology, this updated methodology expands the geographical and temporal coverage of AEAs data, aligns more closely with SEEA emission boundaries, applies the residence principle for air and maritime transport, and improves industry granularity through the systematic and hierarchical use of auxiliary datasets. The methodology retains an “inventory-first” foundation while addressing key limitations of earlier approaches, particularly with respect to international transport and industry allocation.

137. The plausibility checks demonstrated that the methodology can reproduce national CO₂ emission totals, industry structures, and temporal evolution with a high degree of plausibility where input data are consistent and sufficiently detailed, except road transport emissions where significant territory/residence adjustments are necessary. Results also highlight the critical role of data availability in shaping high-quality AEA estimates. To ensure the use of official data sources as much as possible and enhance the applicability of the methodology without relying too much on back-recalculation/extrapolation, comprehensive inventory data are important. PEFA emerges as a key auxiliary dataset for energy-related emission sources to support plausible allocation, especially for obtaining a detailed industry breakdown. Conversely, data gaps—most notably in PEFA coverage—remain a source of uncertainty.

138. Overall, the proposed methodology provides a transparent and replicable framework for generating estimated AEAs that could complement official statistics. It offers a practical basis for improving emissions accounting in countries that do not yet compile full AEAs, and for extending the temporal coverage and improving industry granularity of existing series where official AEA data are available but incomplete. The approach was applied to OECD countries that do not currently report AEAs but can be readily applied to all countries that have the relevant underlying datasets available.

139. Several avenues for the further refinement and extension of this methodology have been identified. These include in order of priority i) the extension of coverage to non-CO₂ GHGs and CO₂ from biomass, ii) territory-residence adjustments for road transport, iii) more granular ISIC breakdown to strengthen the relevance for carbon footprint estimation, iv) improving the allocation of emissions from electricity and heat generation based on the energy source mix, v) disaggregating air and maritime transport emissions beyond transport (ISIC H), and vi) improving the country-specific allocation for off-road vehicles. Moreover, plausibility checks could be further refined to assess more closely the industry structure of emissions.

140. In addition, improvements in inventory inputs are expected in both official and unofficial estimates. Under the Paris Agreement's ETF, Parties are expected to continuously improve the transparency, accuracy, completeness, consistency, and comparability of their GHG inventory reporting over time, while taking into account national capacities and circumstances. Where inventory data gaps persist, the OECD Environment Directorate is developing gap-filling methods to produce more detailed and harmonised emission inventory estimates. Should AEAs be extended to include emissions and removals from Land Use, Land-Use Change and Forestry (LULUCF) in the ongoing update of the SEEA-CF, future methodological work would need to explore how to integrate them into AEA estimates.

141. Furthermore, emissions cross-classified by CRT categories and ISIC/household activities generated in the calculation step for AEA estimates could feed into the development of a single emissions database at the OECD, improving the consistency, granularity, transparency, and accessibility of GHG emissions data.

142. To conclude, the updated methodology presented in this paper improves the availability and usability of AEA estimates by expanding the geographical and temporal coverage, enhancing industry granularity, and strengthening alignment with the SEEA-CF. These advances are achieved using inventory data reported by a wider range of countries under the ETF using the new CRT template, the integration of multiple auxiliary data sources, and the incorporation of experimental estimates for transport emissions. The resulting more complete AEAs are expected to enhance their relevance for analytical applications, including carbon footprint calculations, while improving consistency between GHG emission inventories and AEAs. As data availability continues to evolve (e.g. more complete inventory and PEFA data), the approach set out in this note provides a flexible foundation for further improving the coverage, consistency, and policy relevance of AEAs.

References

- Australia's Department of Climate Change, Energy, the Environment and Water (2025), *National Inventory Report 2023*. [40]
- Australia's Department of Climate Change, Energy, the Environment and Water (2024), *National inventory by economic sector: data tables and methodology 2022*, <https://www.dcceew.gov.au/climate-change/publications/national-greenhouse-accounts-2022/national-inventory-by-economic-sector-data-tables-and-methodology-2022>. [42]
- Cárdenas Rodríguez, M. et al. (2024), "National greenhouse gas emission inventories, filling the gaps in official data", *OECD Environment Working Papers*, No. 252, OECD Publishing, Paris, <https://doi.org/10.1787/f19f1663-en>. [19]
- Citepa (2025), *Méthodologie de l'inventaire – rapport Ominea*, <https://www.citepa.org/donnees-air-climat/methodologie-de-linventaire-omine/>. [43]
- Clarke, D. et al. (2023), "CO2 emissions from global shipping: A new experimental database", *OECD Statistics Working Papers*, No. 2023/04, OECD Publishing, Paris, <https://doi.org/10.1787/bc2f7599-en>. [37]
- Clarke, D. et al. (2022), "CO2 Emissions from air transport: A near-real-time global database for policy analysis", *OECD Statistics Working Papers*, No. 2022/04, OECD Publishing, Paris, <https://doi.org/10.1787/ecc9f16b-en>. [36]
- DANE (2025), *Boletín técnico: Cuenta Ambiental y Económica de Flujos de Materiales de Emisiones al Aire (CAEFM-EA) 2022 – 2023 provisional*, <https://www.dane.gov.co/files/operaciones/CAEFM-EA/bol-CAEFM-EA-2023p.pdf>. [41]
- Environment and Climate Change Canada (2024), *National inventory report 1990-2022: Greenhouse gas sources and sinks in Canada*, <https://unfccc.int/documents/639906>. [44]
- European Union (2011), *Regulation No 691/2011 of the European Parliament and of the Council of 6 July 2011 on European environmental economic accounts*, <https://eur-lex.europa.eu/eli/reg/2011/691/oj>. [5]
- European Union (2010), *Commission Regulation (EU) No 715/2010 of 10 August 2010 amending Council Regulation (EC) No 2223/96 as regards adaptations following the revision of the statistical classification of economic activities NACE Revision 2 and the statistical classification*, <https://eur-lex.europa.eu/legal-content/EN/ALL/?uri=CELEX%3A32010R0715>. [7]

- Eurostat (2025), *National reference metadata, Luxembourg, Air emissions accounts by NACE Rev. 2 activity (env_ac_ainah_r2)*, https://ec.europa.eu/eurostat/cache/metadata/EN/env_ac_ainah_r2_simsae.lu.htm#contactDisseminated. [39]
- Eurostat (2025), *Statistics Explained: Greenhouse gas emission footprints*, https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Greenhouse_gas_emission_footprints. [3]
- Eurostat (2015), *Annex I (Correspondence between CRF/NFR - NACE Rev.2) to Manual for Air emissions Accounts (2015 edition)*, <https://ec.europa.eu/eurostat/documents/1798247/6191529/Annex-I-%28Correspondence-between-CRF-NFR-NACE-Rev.-2%29-to-Manual-for-Air-Emissions-Accounts-%282015-edition%29/>. [45]
- Eurostat (2015), *Manual for air emissions accounts*, <https://ec.europa.eu/eurostat/documents/3859598/7077248/KS-GQ-15-009-EN-N.pdf/ce75a7d2-4f3a-4f04-a4b1-747a6614eeb3?t=1447948110000>. [8]
- Eurostat (2014), *Physical Energy Flow Accounts (PEFA) Manual 2014*, https://seea.un.org/sites/seea.un.org/files/Data_collection/pefa-manual-2014-v20140515.pdf. [14]
- Flachenecker, F., E. Guidetti and P. Pionnier (2018), “Towards global SEEA Air Emission Accounts: Description and evaluation of the OECD methodology to estimate SEEA Air Emission Accounts for CO₂, CH₄ and N₂O in Annex-I countries to the UNFCCC”, *OECD Statistics Working Papers*, No. 2018/11, OECD Publishing, Paris, <https://doi.org/10.1787/7d88dfdd-en>. [1]
- Gütschow, J., D. Busch and M. Pflüger (2025), *The PRIMAP-hist national historical emissions time series (1750-2024) v2.7*. Zenodo, <https://doi.org/10.5281/zenodo.17090760>. [22]
- Gütschow, J. et al. (2016), *The PRIMAP-hist national historical emissions time series*, <https://doi.org/10.5194/essd-8-571-2016>. [25]
- Hishiyama et al. (2025), “汚染調整済経済成長率等の新たな試算について (New Estimates for Pollution-Adjusted Economic Growth Rates, etc.) [in Japanese]”, *ESRI Research Note 94*, https://www.esri.cao.go.jp/jp/esri/archive/e_rnote/e_rnote100/e_rnote094.pdf. [46]
- Horvát, P. and C. Webb (2020), “The OECD STAN Database for industrial analysis: Sources and methods”, *OECD Science, Technology and Industry Working Papers*, No. 2020/10, OECD Publishing, Paris, <https://doi.org/10.1787/ece98fd3-en>. [31]
- IEA (2026), *Greenhouse gas emissions estimates matching the energy module of 2006 IPCC Guidelines*, <https://www.iea.org/data-and-statistics/data-product/greenhouse-gas-emissions-from-energy>. [24]
- IEA (2026), *World extended energy balances*, <https://www.iea.org/data-and-statistics/data-product/world-energy-balances>. [29]
- IEA (2025), *Greenhouse Gas Emissions from Energy 2025 Edition, Database documentation*, https://iea.blob.core.windows.net/assets/f22141ca-b3ed-444a-8157-243732fbf62d/WORLD_GHG_Documentation_2025.pdf. [21]

- IEA (2025), *World Energy Balances, Database Documentation*, [15]
<https://iea.blob.core.windows.net/assets/fddaf908-1716-431b-ab9b-a21f7a6cdd76/WorldEnergyBalancesdocumentationApril2025edition.pdf>.
- IMF (2025), *G20 Data Gaps Initiative - Phase Three (DGI-3) Progress Report, October 2025*, [4]
<https://www.imf.org/-/media/files/news/seminars/dgi/documents/third-phase-of-the-g20-data-gaps-initiative-third-progress-report-2025.pdf>.
- INEGI (2026), *Registered Motor Vehicles in Circulation*, [54]
https://en.www.inegi.org.mx/programas/vehiculosmotor/#tabular_data.
- International Road Federation (2026), *Total four wheeld traffic (by vehicle type)*, [55]
<https://datawarehouse.worldroadstatistics.org/dashboard>.
- IPCC (2019), *2019 Refinement to the 2006 Guidelines for National Greenhouse Gas Inventories, Volume 1, General Guidance and Reporting*, [17]
<https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol1.html>.
- IPCC (2019), *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2 Energy*, [58]
<https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol2.html>.
- IPCC (2006), *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 1 General Guidance and Reporting*, [16]
<https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol1.html>.
- IPCC (2006), *2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2, Energy*, [12]
<https://www.ipcc-nggip.iges.or.jp/public/2006gl/vol2.html>.
- Korea Transportation Safety Authority (2021), *용도별 차종별 연료별 연간 자동차주행거리 (KOR Vehicle Kilometers Travelled)*, [56]
<https://www.data.go.kr/data/15088452/fileData.do>.
- Netherlands Emissieregistratie (2025), *emissieregistratie, datareeks 1990-2023 Definitief*, [18]
<https://www.emissieregistratie.nl/>.
- OECD (2026), *Air emissions - Greenhouse gas emissions Inventories*, [23]
<https://data-explorer.oecd.org/s/4n1>.
- OECD (2026), *Air Emissions Accounts*, [38]
<https://data-explorer.oecd.org/s/4n3>.
- OECD (2026), *Air transport emissions (experimental)*, [35]
<https://data-explorer.oecd.org/s/4n7>.
- OECD (2026), *Physical Energy Flow Accounts: emissions-relevant use*, [27]
<https://data-explorer.oecd.org/s/4n4>.
- OECD (2026), *Physical Energy Flow Accounts: supply by activity*, [32]
<https://data-explorer.oecd.org/s/4y0>.
- OECD (2026), *Physical Energy Flow Accounts: use by activity*, [28]
<https://data-explorer.oecd.org/s/4n6>.
- OECD (2026), *SUT Supply*, [33]
<https://data-explorer.oecd.org/s/4y1>.
- OECD (2025), *Maritime transport CO2 emissions (experimental)*, [34]
<https://data-explorer.oecd.org/s/4za>.

- OECD (2025), *STAN Database for Structural Analysis, 2025 edition*, <https://data-explorer.oecd.org/s/4na>. [30]
- Paolo, F. et al. (2024), "Satellite mapping reveals extensive industrial activity at sea", *Nature*, Vol. 625, pp. 85-91, <https://doi.org/10.1038/s41586-023-06825-8>. [60]
- Pizarro, R. and F. Drogo (Forthcoming), *Methodology to estimate gaps in National Greenhouse Gas Inventories*. [26]
- Sakata, S., A. Aklilu and R. Pizarro (2024), "Greenhouse gas emissions data: Concepts and data availability", *OECD Statistics Working Papers*, No. 2024/03, OECD Publishing, Paris, <https://doi.org/10.1787/b3e6c074-en>. [11]
- Schenau, S. (2025), *SEEA Central Framework 2028 update, Scoping Note, Issue B3 – Treatment of carbon flows in the SEEA CF*, https://seea.un.org/sites/default/files/documents/SEEACF_UPDATE/seeacf_sn_b3_carbon_flows_tc_review.pdf. [59]
- Stadler, K. (2021), *EXIOBASE 3*, <https://doi.org/10.5281/zenodo.5589597>. [51]
- Statistics New Zealand (2020), "Environmental-economic accounts: Sources and methods (third edition)", <https://www.stats.govt.nz/methods/environmental-economic-accounts-sources-and-methods/>. [47]
- UK National Atmospheric Emissions Inventory (2024), *UK Greenhouse Gas Pivot Table*, <https://naei.energysecurity.gov.uk/greenhouse-gases/greenhouse-gas-emissions-data>. [48]
- UK ONS (2025), *Energy use: by industry, source and fuel*, <https://www.ons.gov.uk/economy/environmentalaccounts/datasets/ukenvironmentalaccountsenergyusebyindustrysourceandfuel/current>. [53]
- UK ONS (2024), *Energy use: by industry, source and fuel*, <https://www.ons.gov.uk/economy/environmentalaccounts/datasets/ukenvironmentalaccountsenergyusebyindustrysourceandfuel/current>. [49]
- UN ECLAC (2026), *Repository of supply and use tables and input-output tables in Latin America and the Caribbean*, <https://statistics.cepal.org/repository/cou-mip/index.html?lang=en>. [52]
- UNCEEA (2026), *Results of the 2025 Global Assessment of Environmental-Economic Accounting and Supporting Statistics*, https://unstats.un.org/UNSDWebsite/statcom/session_57/documents/BG-3f-Global_Assessment_2025_UNSC-E.pdf. [6]
- UNFCCC (2023), *Greenhouse Gas Inventory Data - Flexible Queries Annex I Parties*, https://di.unfccc.int/flex_annex1. [57]
- UNFCCC (2021), *Common Reporting Tables (CRT) on National Inventory Reports of anthropogenic emissions by sources and removals by sinks of greenhouse gases*, <https://unfccc.int/documents/311076>. [9]
- UNFCCC (2018), *Decision 18/CMA.1: Modalities, procedures and guidelines for the transparency framework for action and support referred to in Article 13 of the Paris Agreement*, <https://unfccc.int/resource/tet/0/00mpg.pdf>. [20]

- United Nations (2014), *System of Environmental-Economic Accounting 2012 - Central Framework*, <https://seea.un.org/content/seea-central-framework>. [13]
- United Nations (2008), *International Standard Industrial Classification of All Economic Activities, Revision 4*. [10]
- US EPA (2023), *NAICS Crosswalk EPA GHGI*, <https://github.com/USEPA/flowsa/tree/v2.0.0/flowsa/data/activitytosectormapping>. [50]
- Yamano, N., M. Lioussis and A. Cimper (2024), “Measuring greenhouse gas footprints in global production networks: New perspectives on emissions embodied in production chains and final demand patterns”, *OECD Science, Technology and Industry Working Papers*, No. 2024/12, OECD Publishing, Paris, <https://doi.org/10.1787/fc426ab9-en>. [2]

Annex A. Correspondence table

1. Existing correspondence tables mapping UNFCCC CRF/CRT emission categories to ISIC Rev.4 economic activities and households are often limited in scope or consistency. In particular, residual “Other” inventory sub-categories, flexible reporting practices, and the frequent use of “IE (included elsewhere)” notations can blur distinctions between inventory categories and complicate systematic allocation. In addition, some available mappings rely on oversimplified assumptions – for example, assigning all road transport emissions exclusively to land transport activities and households – which can result in implausible industry and household distributions in key emission categories.

2. To address these limitations and enable a more consistent allocation of inventory emissions across countries, the proposed methodology develops a comprehensive reference correspondence table by synthesising information from multiple international and national sources. These include:

- the 2006 IPCC Guidelines for National GHG Inventories, which provide mapping to ISIC Revision 3 economic activities for several fuel combustion subcategories (IPCC, 2006_[16]).²⁷
- Eurostat correspondence tables (Eurostat, 2015_[8])
- the default correspondence table embedded in the IMF’s AEA estimation tool, and
- correspondence tables derived from countries’ detailed underlying data and methodological documentation

Table A.1 reviews available correspondence tables and related datasets that map from UNFCCC inventory emission categories in CRF/CRT to ISIC economic activities and households, either directly or indirectly.

Table A.1. Information related to correspondence between inventory categories and economic and household activities

Country / Organisation	Industry classification systems	Direct alignment with ISIC Rev. 4 (2-digit)	Level of industry detail	Quantified allocation information
Australia	ANZSIC 2006	No	Medium (some categories bundled)	Not available
Canada	NAICS	No	Medium	Available (Emissions data)
France	NACE Rev. 2	Yes	Detailed	Not available
Japan	JSIC	No	Detailed	Not available
Netherlands	SBI	Yes	Detailed	Available
New Zealand	ANZSIC 2006	No	Detailed	Not available
Norway	SIC Norway	Yes	Detailed	Not available
United Kingdom	UK SIC	Yes	Detailed	Available (energy-related categories)
United States	NAICS 2012	No	Detailed	Not available
Eurostat	NACE Rev. 2	Yes	Detailed (See Note)	See Note
IMF	ISIC Rev. 4	Yes	Detailed	Default allocation ratio
IPCC	ISIC Rev. 3	No	Detailed but incomplete	Not available

Note: Acronyms refer to the Australian and New Zealand Standard Industrial Classification (ANZSIC), Japan Standard Industrial Classification (JSIC), North American Industry Classification System (NAICS), Standaard Bedrijfsindeling (SBI), and Standard Industrial Classification (SIC). Eurostat's correspondence table in the Annex to the AEA Manual is detailed, however, some inventory categories are not assigned to specific NACE codes because allocation depends on country-specific practices. The published table does not include quantified allocation coefficients. In practice, Eurostat applies a separate allocation matrix containing allocation ratios when estimating AEAs as the basis for calculating carbon footprints.

Source: (Australia's Department of Climate Change, Energy, the Environment and Water, 2024^[42]; Citepa, 2025^[43]; Environment and Climate Change Canada, 2024^[44]; Eurostat, 2015^[45]; Hishiyama et al., 2025^[46]; IPCC, 2006^[16]; Netherlands Emissieregistratie, 2025^[18]; Statistics New Zealand, 2020^[47]; UK National Atmospheric Emissions Inventory, 2024^[48]; UK ONS, 2024^[49]), (US EPA, 2023^[50]).

3. When multiple correspondence tables suggest different industry allocations for the same CRF/CRT category, priority is given to mappings grounded in the IPCC Guidelines. In particular, the description of emission source categories and their mapping to ISIC Rev.3 divisions are used as a primary reference and combined with their correspondence to ISIC Rev.4.

4. Where IPCC-based mappings are not available, priority is given to mappings supported by quantitative empirical evidence, such as those derived from detailed national data for the Netherlands and the United Kingdom, over indicative mappings without such evidence. Industry allocations suggested by only a limited number of correspondence examples are generally excluded from the OECD reference correspondence table, as they may reflect country-specific secondary activities and are not necessarily generalisable.

5. Among the reviewed sources, the correspondence tables derived from the Netherlands and the United Kingdom's data are particularly informative. Both provide high industry granularity, alignment with ISIC classifications, and detailed empirical data that enable analysis of the quantified distribution of emissions across economic activities.

6. The Netherlands' emissions database classifies emissions from stationary sources using detailed national source categories that map to CRF emission codes. A key strength of this system is that many of

these source categories are also linked to SBI codes, the Dutch industry classification system, which is compatible with ISIC at the 2-digit division level. This coding system enables cross-classification of emissions by CRF category and ISIC-equivalent industry, allowing empirical validation of allocation patterns and assessment of how emissions from specific inventory categories are distributed across economic activities in practice.

7. The United Kingdom's national dissemination of inventory data provides a similarly strong empirical basis. The UK inventory includes source and fuel-product categories that are more detailed than those reported in standard CRT/CRF submissions. These detailed categories are also used to disaggregate energy-use data by economic activity, allowing emissions, fuel types, and industries to be consistently linked. By combining the detailed inventory and industry energy-use data, the OECD was able to replicate the UK's CO₂ AEA with a total misallocation ratio of only 1.1% at the detailed A*64 industry level. This strong empirical correspondence supports the use of UK-derived allocation ratios in the proposed methodology for selected cases, such as allocating emissions from 1.A.2.g.vii Off-road vehicles and other machinery, where no reliable country-specific data are otherwise available.

8. Synthesising the evidence from these sources, Table A.2 presents the OECD reference correspondence table used in this paper to allocate emissions from UNFCCC CRT categories to ISIC Rev.4 activities and households consistently across countries. The table may be updated as further methodological developments and empirical evidence become available. For example, future refinements could include improved allocation of CRT sub-categories under 1.B Fugitive emissions from fuels and a finer allocation of CRT categories related to the chemical industry, allowing a distinction between ISIC C20 (Chemical products) and C21 (Pharmaceutical products).

Table A.2. Reference Correspondence Table for CO₂ emissions

IPCC	Description	ISIC and households
1.A.1.a	Public Electricity and Heat Production	D, E37T39
1.A.1.a.i	Electricity Generation	D, E37T39
1.A.1.a.ii	Combined Heat and Power Generation	D, E37T39
1.A.1.a.iii	Heat Plants	D, E37T39
1.A.1.a.iv	Other	D, E37T39
1.A.1.b	Petroleum Refining	C19
1.A.1.c.i	Manufacture of Solid Fuels	C19
1.A.1.c.ii	Oil and Gas Extraction	B
1.A.1.c.iii	Other Energy Industries	D
1.A.1.c.iv	Other	C19
1.A.2.a	Iron and Steel	C24
1.A.2.b	Non-Ferrous Metals	C24
1.A.2.c	Chemicals	C20
1.A.2.d	Pulp, Paper and Print	C17, C18
1.A.2.e	Food Processing, Beverages and Tobacco	C10T12
1.A.2.f	Non-metallic Minerals	C23
1.A.2.g.i	Manufacturing of Machinery	C25, C26, C27, C28
1.A.2.g.ii	Manufacturing of Transport Equipment	C29, C30
1.A.2.g.iii	Mining (Excluding Fuels) and Quarrying	B
1.A.2.g.iv	Wood and Wood Products	C16
1.A.2.g.v	Construction	F
1.A.2.g.vi	Textile and Leather	C13T15
1.A.2.g.vii	Off-road Vehicles and Other Machinery	B, C10T12, C13T15, C16, C19, C21, C22, C25, C27, C28, C29, C30, C31_32, C33, E36, E37T39, F, H49, H52
1.A.2.g.viii	Other	C22, C31_32, C33
1.A.3.b	Road Transportation	All industries and HH_TRA
1.A.3.b.i	Cars	All industries and HH_TRA
1.A.3.b.ii	Light Duty Trucks	All industries and HH_TRA
1.A.3.b.iii	Heavy Duty Trucks and Buses	All industries
1.A.3.b.iv	Motorcycles	HH_TRA
1.A.3.b.v	Other	All industries and HH_TRA
1.A.3.c	Railways	H49
1.A.3.e.i	Pipeline Transport	H49
1.A.3.e.ii	Other	H52
1.A.4.a	Commercial/Institutional	G45, G46, G47, I, J58, J59_60, J61, J62_63, K64, K65, K66, L, M69_70, M71, M72, M73, M74_75, N77, N78, N79, N80T82, O, P, Q86, Q87_88, R90T92, R93, S94, S95, S96, T
1.A.4.b	Residential	HH_HEAT
1.A.4.b.i	Stationary Combustion	HH_HEAT
1.A.4.b.ii	Off-road Vehicles and Other Machinery	HH_OTH
1.A.4.b.iii	Other	HH_OTH
1.A.4.c	Agriculture/Forestry/Fishing	A01, A02, A03

1.A.4.c.i	Stationary	A01
1.A.4.c.ii	Off-road Vehicles and Other Machinery	A01, A02
1.A.5	Other (Not specified elsewhere)	All industries and HH_OTH
1.A.5.a	Stationary	All industries and HH_OTH
1.A.5.b	Mobile	O
1.B.1	Solid Fuels	B
1.B.1.a	Coal Mining and Handling	B
1.B.1.b	Solid Fuel Transformation	C19
1.B.1.c	Other	B
1.B.2	Oil and Natural Gas and Other Emissions from Energy Production	B
1.B.2.a	Oil	B
1.B.2.a.i	Exploration	B
1.B.2.a.ii	Production	B
1.B.2.a.iii	Transport	H49
1.B.2.a.iv	Refining / Storage	C19, H52
1.B.2.a.v	Distribution of Oil Products	G47
1.B.2.a.vi	Other	B
1.B.2.b	Natural Gas	B
1.B.2.b.i	Exploration	B
1.B.2.b.ii	Production	B
1.B.2.b.iii	Processing	B
1.B.2.b.iv	Transmission and Storage	H49
1.B.2.b.v	Distribution	D
1.B.2.b.vi	Other	B
1.B.2.c.i	Venting	B
1.B.2.c.i.1	Oil	B
1.B.2.c.i.2	Gas	B
1.B.2.c.i.3	Combined	B
1.B.2.c.ii	Flaring	B
1.B.2.c.ii.1	Oil	B
1.B.2.c.ii.2	Gas	B
1.B.2.c.ii.3	Combined	B
1.B.2.d	Other	B
1.C.1	Transport of CO ₂	H49, H50
1.C.1.a	Pipelines	H49
1.C.1.b	Ships	H50
1.C.1.c	Other	H49
1.C.2	Injection and Storage	B
2.A	Mineral Industry	C23
2.A.1	Cement Production	C23
2.A.2	Lime Production	C23
2.A.3	Glass production	C23
2.A.4	Other Process Uses of Carbonates	C20, C23, C24, D, F

2.A.4.a	Ceramics	C23
2.A.4.b	Other Uses of Soda Ash	C20
2.A.4.c	Non-metallurgical Magnesium Production	C23
2.A.4.d	Other	C24, D, F
2.B	Chemical Industry	C20
2.C	Metal Industry	C24
2.D	Non-energy Products from Fuels and Solvent Use	All industries and HH_TRA
2.D.1	Lubricant Use	All industries and HH_TRA
2.D.2	Paraffin Wax Use	HH_OTH
2.D.3.b	Road Paving with Asphalt	F
2.D.3.c	Asphalt Roofing	F
2.G	Other Product Manufacture and Use	C10TC12, C13T15, C16, C17, C18, C20, C22, C23, C24, C25, C27, C28, C29, C30, C31_32, C33
2.H	Other	C10T12, C17
2.H.1	Pulp and Paper	C17
2.H.2	Food and Beverages Industry	C10T12
2.H.3	Other	C10T12, C17
3	Agriculture	A01
5	Waste	E37T39

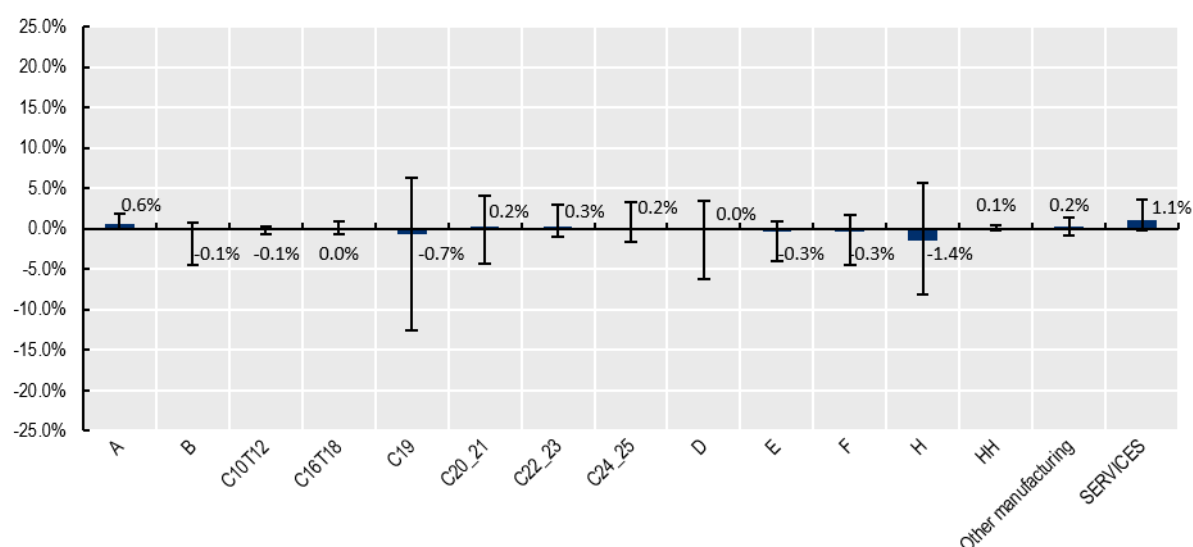
Note: Categories 1.A.3.a Domestic aviation, 1.A.3.d Domestic Navigation, and 1.A.4.c.iii Fishing are excluded because they are not used for allocation. LULUCF is excluded to align with the AEA emission scope under the current SEEA-CF. CRT categories that are irrelevant to CO₂ emissions (e.g. "2.F Product Uses as Substitutes for ODS") are not included. Parent inventory categories do not necessarily map to all the economic activities used for allocating their child inventory categories, because, in some cases, allocation to certain economic activities is appropriate only when emissions are reported at a more detailed level. A01: Crop and animal production, hunting and related service activities, A02: Forestry and logging, A03: Fishing and aquaculture, B: Mining and quarrying, C10T12: Manufacture of food products; beverages and tobacco products, C13T15: Manufacture of textiles, wearing apparel, leather and related products, C16: Manufacture of wood and of products of wood and cork, except furniture; manufacture of articles of straw and plaiting materials, C17: Manufacture of paper and paper products, C18: Printing and reproduction of recorded media, C19: Manufacture of coke and refined petroleum products, C20: Manufacture of chemicals and chemical products, C21: Manufacture of basic pharmaceutical products and pharmaceutical preparations, C22: Manufacture of rubber and plastics products, C23: Manufacture of other non-metallic mineral products, C24: Manufacture of basic metals, C25: Manufacture of fabricated metal products, except machinery and equipment, C26: Manufacture of computer, electronic and optical products, C27: Manufacture of electrical equipment, C28: Manufacture of machinery and equipment n.e.c., C29: Manufacture of motor vehicles, trailers and semi-trailers, C30: Manufacture of other transport equipment, C31_32: Manufacture of furniture; other manufacturing, C33: Repair and installation of machinery and equipment, D: Electricity, gas, steam and air conditioning supply, E36: Water collection, treatment and supply, E37T39: Sewerage, waste management, remediation activities, F: Construction, G45: Wholesale and retail trade and repair of motor vehicles and motorcycles, G46: Wholesale trade, except of motor vehicles and motorcycles, G47: Retail trade, except of motor vehicles and motorcycles, H49: Land transport and transport via pipelines, H50: Water transport, H51: Air transport, H52: Warehousing and support activities for transportation, H53: Postal and courier activities, HH_HEAT: Heating and cooling activities by households, HH_OTH: Other activities by households, HH_TR: Transport activities by households, I: Accommodation and food service activities, J58: Publishing activities, J59_60: Motion picture, video, television programme production; programming and broadcasting activities, J61: Telecommunications, J62_63: Computer programming, consultancy, and information service activities, K64: Financial service activities, except insurance and pension funding, K65: Insurance, reinsurance and pension funding, except compulsory social security, K66: Activities auxiliary to financial service and insurance activities, L68: Real estate activities, M69_70: Legal and accounting activities; activities of head offices; management consultancy activities, M71: Architectural and engineering activities; technical testing and analysis, M72: Scientific research and development, M73: Advertising and market research, M74_75: Other professional, scientific and technical activities; veterinary activities, N77: Rental and leasing activities, N78: Employment activities, N79: Travel agency, tour operator, reservation service and related activities, N80T82: Security and investigation, service and landscape, office administrative and support activities, O: Public administration and defence; compulsory social security, P: Education, Q86: Human health activities, Q87_88: Residential care activities and social work activities without accommodation, R90T92: Creative, arts and entertainment activities; libraries, archives, museums and other cultural activities; gambling and betting activities, R93: Sports activities and amusement and recreation activities, S94: Activities of membership organizations, S95: Repair of computers and personal and household goods, S96: Other personal service activities, T: Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use. Section U, Activities of extraterritorial organisations and bodies, is not used for allocation.

Annex B. Distribution of misallocated emissions excluding air transport, maritime/fishing, and road transport

1. Figure 14 presents the industry and household distribution of misallocated emissions. Large-emitting industries, such as ISIC C19 “Manufacture of coke and refined petroleum products”, ISIC D “Electricity, gas, steam and air conditioning supply”, and ISIC H “Transportation and storage”, exhibit higher misallocation shares. This partly reflects the design of the metric, as misallocation is expressed as a share of total emissions; potential under-estimation in smaller industries is therefore limited in absolute terms and contributes less to the overall misallocation share.
2. In constructing the comparison scope, official AEAs are adjusted by subtracting road transport emissions from each industry and the household sector, subtracting international and domestic air/maritime transport by residents from ISIC H using relevant bridging items. In some countries, however, other industries (e.g. ISIC B “Mining and quarrying”) may account for non-negligible transport-related emissions, such as vessels operated for offshore oil extraction.²⁸ Adjusting transport emissions exclusively within ISIC H may therefore distort industry comparisons and contribute to misallocation.
3. Among large-emitting categories, only a small share of misallocation is attributed to households with very little variation. This suggests that the allocation of CRT 1.A.4.b “Residential” to households broadly reflects official AEA compilation.
4. Smaller CO₂-emitting industries generally exhibit correspondingly smaller misallocation shares. However, some industries, such as ISIC B, E and F, display relatively large variation in misallocation compared to the size of industry emissions.
5. Potential misallocation in ISIC B may partly reflect the adjustment of maritime transport emissions exclusively within ISIC H, as discussed above.
6. Underestimation of CO₂ emissions in ISIC C19 may partly result from the simplified allocation of emissions from sub-categories under CRT 1.B Fugitive emissions from fuels. While the 1-to-1 correspondence between CRT 1.A.1.b Petroleum Refining and ISIC C19 is relatively straightforward, the allocation of emissions from sub-categories under CRT 1.B categories is more complex and may warrant further refinement. This is particularly relevant when the methodology is extended to cover methane emissions, for which fugitive emissions are a key source.
7. Underestimation of CO₂ emissions in ISIC E may be linked to allocation from CRT 1.A.1.a “Public electricity and heat production”. The methodology allocates emissions between ISIC D and E based on physical supply of electricity and heat (in terajoules). In many cases, emissions are allocated entirely to ISIC D. Where a split is applied, however, differences in emission intensity of electricity generation between ISIC D and E—particularly when ISIC D relies more heavily on nuclear or renewable energy—may generate industry discrepancies, which warrant further investigation and methodological refinement.
8. ISIC F is closely linked to emissions from CRT 1.A.2.g.v “Construction” and 1.A.2.g.vii “Off-road vehicles and other machinery”. These five-digit categories under 1.A.2 “Manufacturing industries and

construction” are not always separately reported in official inventories and are supplemented, where necessary, using corresponding categories from the IEA dataset “Greenhouse gas emissions estimates matching the energy module of 2006 IPCC Guidelines”. In addition, allocation of 1.A.2.g.vii currently relies on a coefficient derived from detailed data for a single country due to the lack of internationally comparable off-road energy-use data. These assumptions and data limitations may explain the relatively large misallocations observed for construction in Figure 14 compared with its emissions share in Figure 4.

Figure 14. Industry and household distribution of misallocated emissions (excluding air transport, maritime/fishing and road transport), Country Group 1



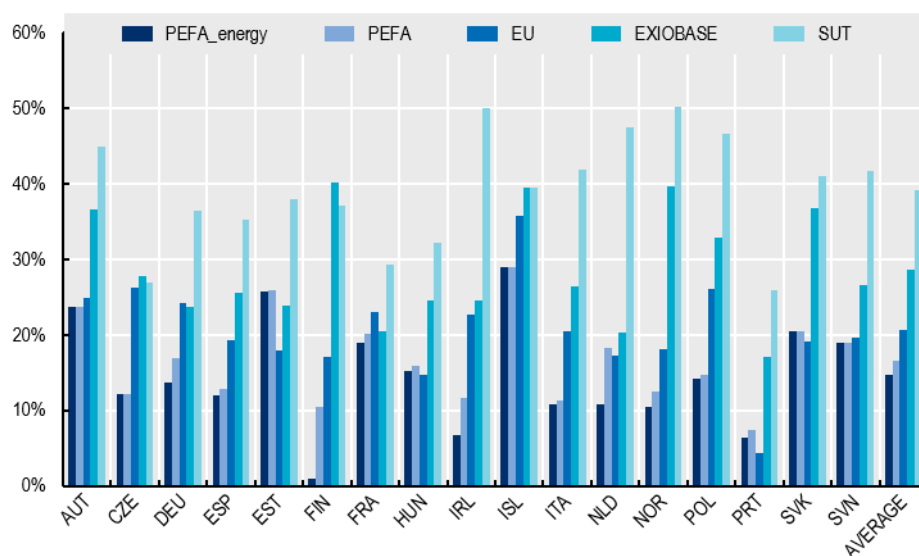
Note: Emissions are allocated following the preferred auxiliary data sources (i.e. PEFA emission-relevant energy use for Energy and output for non-energy). Fishing emissions are excluded for countries that report 1.A.4.c.iii “Fishing” separately. The industry breakdown follows the medium breakdown. Across countries, the share of overall CO₂ emissions which are misallocated to the agriculture, forestry and fishing industry is 0.6% on average. In 88% of the cases, this share lies between -0.1% and +1.9%.

Source: OECD calculation

Annex C. Plausibility of allocation of road transport emissions

- Figure 15 reports the misallocation share across countries. Consistent with the large discrepancies observed in total road transport emissions (Figure 7), misallocation shares are substantial for several countries. In Austria, the misallocation share is close to half of the MAPE of total emissions, reflecting the fact that the overallocation observed at the total level is concentrated in a single industry (ISIC H “Transport”).
- Using the PEFA emission-relevant use table, the misallocation share is very small for Finland (1.1%) and relatively small for Portugal and Ireland (6.5% and 6.8%, respectively).

Figure 15. Misallocation share of road transport CO₂ emissions, Country Group 1

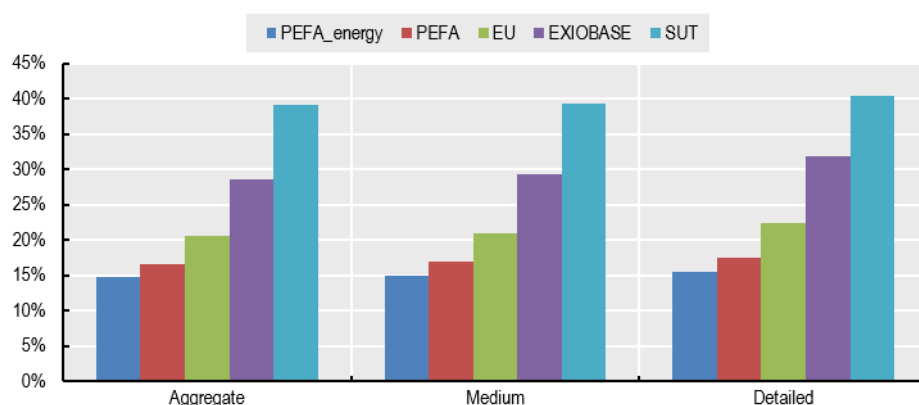


Note: Analysis on the aggregate breakdown level. Excludes Luxembourg (outlier). PEFA_energy stands for PEFA emission-relevant energy use and PEFA for total energy use. PEFA energy products P14 Motor spirit and P17 Transport diesel provide the basis for calculating allocation coefficients. EU stands for the average allocation coefficients from European countries
Source: OECD calculation

- Figure 16 presents misallocation shares using different auxiliary data sources. Unlike the analysis excluding air transport, maritime/fishing, and road transport emissions, the distinction between PEFA tables (emission-relevant energy use and total use) leads to noticeable differences in misallocation shares. Empirical evidence suggests that the gap between emission-relevant use and total use of motor spirit in ISIC C19 “Manufacture of coke and refined petroleum products” and C20 “Manufacture of chemicals and chemical products” is substantial in some countries (OECD, 2026^[27]; OECD, 2026^[28]), possibly reflecting the use of motor spirit as a chemical feedstock or solvent.

4. The allocation based on the European simple average is calculated by excluding the country itself in order to avoid bias (for example, the average across all countries reporting road-transport AEAs, except Austria, is applied when allocating road transport emissions in Austria). The allocation based on the European average produces lower misallocation shares than those obtained using the unofficial SUTs from EXIOBASE. Similar results are observed for non-European countries in Group 2 that do not compile official PEFA data (i.e. Korea and the United States).
5. Using aggregated petroleum products from official SUTs results in the highest misallocation shares among the auxiliary data sources considered in this benchmark. Because this aggregate product category includes oil products other than gasoline and diesel, it does not provide a plausible basis for allocating road transport emissions.
6. Misallocation shares within the same auxiliary data do not vary substantially across different levels of industry and household breakdown. This likely reflects the fact that ISIC H “Transportation and storage” and households—which account for the largest share of road transport emissions—are not affected by these industry and household disaggregations.

Figure 16. Misallocation shares, across different auxiliary data



Note: Simple average is calculated excluding Luxembourg (outlier). PEFA_energy stands for PEFA emission-relevant energy use and PEFA for total energy use. PEFA energy products P14 Motor spirit and P17 Transport diesel provide the basis for calculating allocation coefficients. EU stands for the average allocation coefficients from European countries.

Source: OECD calculations using allocation coefficients derived from (OECD, 2026^[27]) (OECD, 2026^[28]) (OECD, 2026^[38]) (Stadler, 2021^[51])

7. Figure 17 presents the industry and household distribution of misallocated road transport emissions. Misallocation may arise from two main sources: (i) the simplifying assumption regarding territory–residence adjustments, and (ii) the allocation coefficients used to distribute emissions across economic activities and households.
8. ISIC H and households are most directly affected by territory–residence adjustments. For example, road transport emissions from non-resident transport companies fuelling on the territory are included in the OECD estimates, whereas they are deducted in official AEAs. This may contribute to overestimation of emissions in the transport industry. Conversely, road transport emissions from resident tourists fuelling abroad are excluded from OECD estimates, potentially leading to underestimation of household road transport emissions.
9. Misallocation may also arise from the allocation coefficients derived from PEFA energy products P14 “Motor gasoline (without bio)” and P17 “Transport diesel (without bio)”. Diesel fuel is used not only for

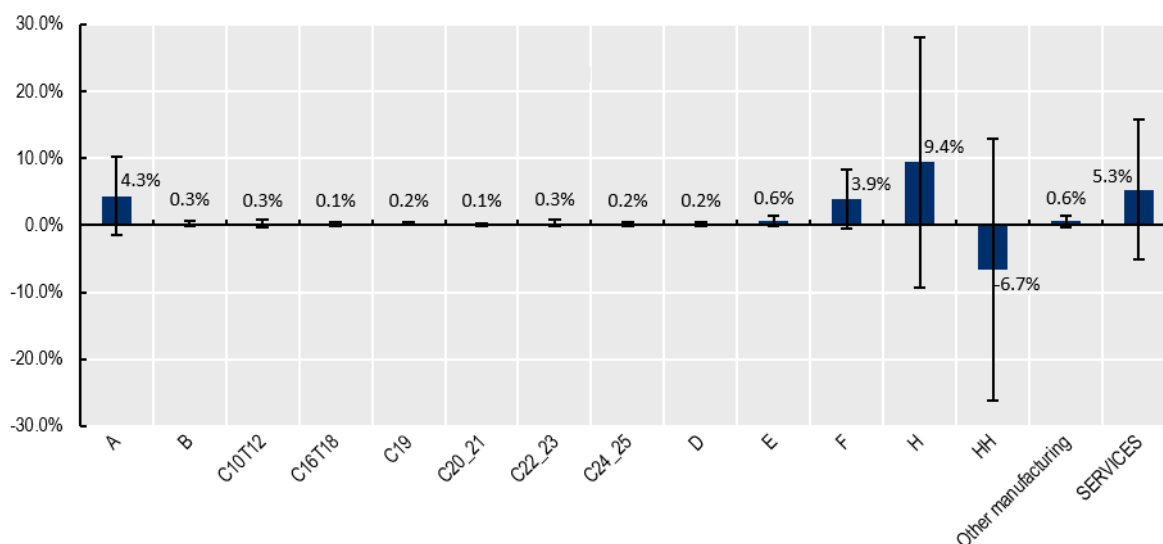
on-road transport but also for off-road purposes, such as agricultural tractors, forklifts in manufacturing facilities or wholesale and retail warehouses, and construction machinery (e.g. bulldozers and cranes).

10. Fuels under PEFA energy products P17 “Transport diesel (without bio)” and P18 “Heating and other gasoil (without bio)” may be chemically identical and therefore not always clearly separable in practice. If P17 includes diesel used for off-road vehicles, this may lead to an overallocation of road transport emissions to industries such as ISIC A “Agriculture, forestry and fishing”, ISIC C “Manufacturing”, ISIC F “Construction”, and ISIC G “Wholesale and retail trade”. In addition, CRT categories 1.A.4.c “Agriculture/Forestry/Fishing”, 1.A.2 “Manufacturing Industries and Construction”, and 1.A.4.a “Commercial/Institutional” each include a subcategory for “Off-road Vehicles and Other Machinery”, while it is not uncommon that these categories are reported blank or as “IE (included elsewhere)” in CRT reporting. As a result, when these CRT off-road categories are reported separately, vehicle-related emissions may be counted both through allocation from off-road CRT categories and through road transport emissions, potentially leading to an overestimation of emissions in these industries. This issue may warrant further methodological refinement.

11. Similarly, PEFA energy product P14 “Motor gasoline (without bio)” is primarily associated with road vehicles but may also be used for other purposes such as gasoline-powered generators, potentially contributing to overallocation of emissions to the corresponding industries.

12. Given the large total-level discrepancies in road transport emissions driven by territory–residence adjustments, the misallocation share—expressed as a proportion of total emissions—tends to be elevated in several countries. Further analysis of industry emission shares in official and estimated AEAs would help disentangle total-level effects from structural allocation differences. In addition, incorporating country-specific data on cross-border road traffic—ideally distinguishing between industries and households—would improve the implementation of territory–residence adjustments for road transport emissions.

Figure 17. Industry and household distribution of misallocated road transport emissions, Country Group 1



Note: Simple average excluding Luxembourg (outlier). Emissions are allocated following the preferred auxiliary data sources (i.e. PEFA emission-relevant energy use). The industry breakdown follows the medium breakdown. Across countries, the share of overall CO₂ emissions which are misallocated to the agriculture, forestry and fishing industry is 4.3% on average. In 88% of the cases, this share lies between -1.5% and +21.2%.

Source: OECD Calculation

Notes

¹ Note that the Paris Agreement does not use the distinction between ‘Annex I’ and ‘non-Annex I’ Parties. This paper uses, where necessary, this terminology. For example, the 2018 methodology for estimating AEAs relied on the Common Reporting Format (CRF) which was used by Annex I countries, predating the reporting requirements introduced by the Enhanced Transparency Framework under the Paris Agreement.

² Road transport allocation keys were derived from Denmark, France and Sweden.

³ Greenhouse gas (GHG) emission inventories are compiled in accordance with the guidelines of the Intergovernmental Panel on Climate Change (IPCC) and follow the territory-based principle. National emission inventories are reported by parties to the United Nations Framework Convention on Climate Change (UNFCCC), which has 198 Parties (i.e. countries and the European Union) as of May 2026. Gas and time coverage, as well as the level of detail in emission source breakdowns vary across countries due to flexibility provisions for developing country parties. Developed countries’ inventories typically cover carbon dioxide (CO₂), methane (CH₄), nitrous oxide (N₂O), hydrofluorocarbons (HFCs), perfluorocarbons (PFCs), sulphur hexafluoride (SF₆) and Nitrogen trifluoride (NF₃), and are generally compiled from 1990 to year *t*-2. Emission sources are classified according to IPCC source categories (see also Box 1).

⁴ For example, a country may report numerical values only for 1.A.4.c.i “Stationary” under 1.A.4.c “Agriculture/Forestry/Fishing”, while indicating “IE (included elsewhere)” for 1.A.4.c.ii “Off-road Vehicles and Other Machinery” and 1.A.4.c.iii “Fishing” in its national emission inventory. In such cases, an allocation based strictly on the most detailed (5-digit) level would assign emissions only to ISIC A01 “Crop and animal production, hunting and related service activities”, with no emissions attributed to A02 “Forestry and logging” or A03 “Fishing and aquaculture”. However, if the country has production activities in forestry or fishing activities, assigning zero emissions to these industries would be implausible.

⁵ For example:

$$\begin{bmatrix} 0.6 & 1 & 0 \\ 0.4 & 0 & 0.2 \\ 0 & 0 & 0.8 \end{bmatrix} \begin{bmatrix} 100 \\ 70 \\ 50 \end{bmatrix} = \begin{bmatrix} 130 \\ 50 \\ 40 \end{bmatrix}$$

This simplified example assumes three conceptual inventory categories (1, 2, and 3) and three conceptual economic categories (A, B, and C). The inventory vector on the left-hand side shows emissions of 100, 70, and 50 tonnes for categories 1, 2, and 3, respectively. Each column of the weight matrix specifies how emissions from a given inventory category are distributed across economic categories.

For instance, the first column of the weight matrix represents a “1-to-many” mapping, in which 60% of emissions from inventory category 1 are allocated to economic category A and 40% to category B. The second column represents a “1-to-1” mapping, in which all emissions from inventory category 2 are allocated to economic category A.

Multiplying the weight matrix by the inventory emissions vector yields the vector on the right-hand side, which shows emissions allocated to economic activities. For example, emissions attributed to economic category A are calculated as the product of the first row of the weight matrix and the inventory vector (60% of 100 tonnes + 100% of 70 tonnes + 0% of 50 tonnes = 130 tonnes).

⁶ Emission data based on (UNFCCC, 2023^[57]). Annex I countries of UNFCCC are Australia, Austria, Belarus, Belgium, Bulgaria, Canada, Croatia, Cyprus, Czechia, Denmark, Estonia, European Union, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Japan, Latvia, Liechtenstein, Lithuania, Luxembourg, Malta, Monaco, Netherlands, New Zealand, Norway, Poland, Portugal, Romania, Russian Federation, Slovakia, Slovenia, Spain, Sweden, Switzerland, Türkiye, Ukraine, United Kingdom of Great Britain and Northern Ireland, United States of America as of 19 February 2026 (https://unfccc.int/process/parties-non-party-stakeholders/parties-convention-and-observer-states?field_national_communications_target_id%5B515%5D=515). Six OECD member countries do not belong to the UNFCCC Annex I: Chile, Colombia, Costa Rica, Israel, Korea, and Mexico.

⁷ For a limited number of cases where key child categories under the same parent are suppressed for confidentiality (e.g. 1.A.1.a “Public electricity and heat production” and 1.A.1.b “Petroleum refining”), allocation cannot be performed. Allocating emissions from their parent category (e.g. “1.A.1 Energy industries”) to the corresponding industries (e.g. ISIC D “Electricity, gas, steam and air conditioning supply” and C19 “Manufacture of coke and refined petroleum products”) would effectively amount to estimating the confidential values of 1.A.1.a and 1.A.1.b, which is beyond the scope of the proposed methodology.

⁸ The 2019 Refinement to the 2006 IPCC Guidelines indicates no refinement for these emission factors (IPCC, 2019^[58]).

⁹ This approach assumes equal emission intensity per unit of electricity or heat supplied across ISIC D and ISIC E37–E39, which could be refined in future work. In practice, auxiliary data often indicate that emissions from CRT 1.A.1.a are allocated entirely to ISIC D, limiting the impact of potential differences in emission intensities between ISIC D and ISIC E37–E39.

¹⁰ If inventory emissions data are available for child categories, such as 1.A.1.a.i Electricity generation or 1.A.1.a.iii Heat plants, only the relevant PEFA product category (P26 or P27, respectively) is used for the allocation.

¹¹ The 1-to-1 mapping from CRT 1.A.3.b.iv Motorcycles to households is a simplifying assumption. There is no global dataset providing motorcycle gasoline use by ISIC activity and households. Applying allocation percentages derived from total gasoline use across industries and households would reflect predominantly car and truck use, thereby underestimating households motorcycle emission and overestimating those attributed to industries. Detailed UK energy data indicate that households account for 99% to 100% of petrol use for motorcycles (UK ONS, 2025^[53]). In the European countries assessed in the 2018 methodology, households account for more than 95% of emissions from mopeds and motorcycles in France and Denmark and around 80% in Sweden (Flachenecker, Guidetti and Pionnier, 2018^[11]). Moreover, CO₂ emissions from motorcycles account for approximately 0.15% of total emissions and 0.7% of road transport emissions for Annex I countries, suggesting that the impact of this assumption on overall allocation results is negligible.

¹² Fuel breakdowns such as gasoline and diesel oil in the CRT are reported within the road transportation category and therefore refer specifically to transport-related fuel use. Other fuel categories in CRT (i.e. “liquefied petroleum gases (LPG)”, “other liquid fuels”, “gaseous fuels”, and “other fossil fuels”) account for a minor share of total road transport emissions. Road transport emissions are therefore divided only between “diesel oil” and “gasoline” using the shares of these two fuel types (i.e. the sum of adjusted emissions from diesel oil and gasoline matches the category total).

¹³ If P17 is not reported, allocation is performed using the allocation coefficient derived only from P14.

¹⁴ For instance, in case of Latin American countries UN ECLAC (UN ECLAC, 2026^[52]).

¹⁵ The subcategory “Off-road vehicles and other machinery” also appears under 1.A.4.a “Commercial/Institutional”, 1.A.4.b “Residential”, and 1.A.4.c “Agriculture/Forestry/Fishing”. For 1.A.4.a.ii, emissions from Off-road vehicles and other machinery are allocated across the relevant service industries together with the other subcategories under 1.A.4.a, following the standard hierarchy of auxiliary data. Emissions under 1.A.4.b.ii are allocated entirely to “Households, other” (1:1 mapping). Emissions under 1.A.4.c.ii are split between ISIC A01 and A02 using the standard hierarchy of auxiliary data.

¹⁶ This may introduce a slight error in case of aviation and water activities (e.g. private jets or speedboats) undertaken by activities outside of transport, however this error is expected to be quite small. In addition, emissions figures for fishing can be considered a lower-bound estimate, as a large share of industrial fishing vessels are not captured by publicly available Automatic Identification System (AIS) data (Paolo et al., 2024^[60]). For countries that do not separately report CRT 1.A.4.c.iii. “Fishing”, emissions are allocated from the parent category 1.A.4.c “Agriculture/Forestry/Fishing” using the standard hierarchy of auxiliary data. In this case, only emissions from vessels excluding fishing are taken from the OECD maritime transport emissions database to avoid potential double-counting.

¹⁷ The 2018 OECD methodology for estimating AEAs used the CRF as the primary source of inventory data. Consequently, the coverage of the 2018 approach was restricted to the Annex I countries.

¹⁸ As of February 2026, not all CRT subcategories under CRT 2 IPPU have been extracted by the OECD, but work is ongoing. When all data have been extracted, the calculations will be updated.

¹⁹ CRT submissions are available from https://unfccc.int/reports?f%5B0%5D=document_type%3A4593

²⁰ National dissemination with longer time series of PEFA is available for Australia, Colombia, and Costa Rica from the following links. <https://www.abs.gov.au/statistics/industry/energy/energy-account-australia> <https://www.dane.gov.co/index.php/estadisticas-por-tema/ambientales/cuenta-satelite-ambiental-csa#cuenta-ambiental-y-economica-de-flujos-de-energia-cae-fe> and <https://www.bccr.fi.cr/en/Economic-Indicators/environmental-accounts>

²¹ Contribution-based methods are not used, unlike the extrapolation of CRT data, as alignment between PEFA and Energy Balances totals is less certain due to differences in accounting principles and the non-identical mapping of products and industries/flows. Under these conditions, changes observed in Energy Balances may not directly correspond to changes in PEFA. Aggregate total growth rates are not applied, as the extrapolated PEFA data are used solely to derive industry allocation coefficients, which require consistency at the industry level rather than at the aggregate level.

²² Examples of country-specific road transport statistics that could support such refinements include vehicle-kilometre-travelled data for Korea (Korea Transportation Safety Authority, 2021^[56]), vehicle registration statistics for Mexico (INEGI, 2026^[54]), and traffic statistics compiled by the International Road Federation (International Road Federation, 2026^[55]).

²³ ISIC U “Activities of extraterritorial organizations and bodies” such as activities of international organisations and diplomatic missions, is considered negligible in terms of CO₂ emissions and is not included in the correspondence table presented in Annex A. Therefore, the effective number of industries is 63.

²⁴ Note that possible inclusion of LULUCF in AEAs is currently discussed under the update of the SEEA Central Framework (Schenau, 2025^[59]).

²⁵ The 2019 Refinement to the 2006 IPCC Guidelines indicates no refinement for the uncertainty range related to the emission factors for CO₂ (IPCC, 2019^[17]).

²⁶ With a few exceptions including Israel which reports inventory emissions for consecutive years only from 2003 and Sweden which reports confidential values in the key CRT categories such as 1.A.1.b “Petroleum refining” for some years.

²⁷ The 2019 Refinement to the 2006 IPCC Guidelines does not introduce any updates to the correspondence between IPCC categories and ISIC economic activities (IPCC, 2019^[17]).

²⁸ ISIC B accounts for about half of energy use of coastal shipping based on the UK’s energy data (UK ONS, 2025^[53]).